Recovering Preferences from Mistakes: An Auxiliary Task Approach*

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Abstract

Risk preferences recovered from lottery valuation data are not robust to unverifiable assumptions about the structure of mistakes in the valuations. To address this, we develop a novel approach utilizing Oprea's (2024) deterministic mirrors – deterministic payments that preserve key structural features of lotteries – to correct for the mistakes in lottery valuations. The corrected valuations are closer to risk neutrality than raw valuations; when they deviate from risk neutrality, they are predominantly risk averse. The corrected valuations are more aligned with expected utility theory, in contrast to the raw valuations which exhibit strong probability weighting. Our approach offers a generalizable framework for preference recovery: researchers can use auxiliary tasks with known correct answers to discipline assumptions about mistakes.

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If one makes the wrong assumptions about the stochastic structure of the noise, then one usually makes wrong inferences from the data. (Hey, 2005)

1 Introduction

Risk preferences, when interpreted as a welfare-relevant ordering of alternatives, are important in designing and evaluating economic policies. These preferences are typically recovered by applying a revealed preference argument to observed choices under risk. However, one caveat of applying the recovered preferences for welfare analysis is the possibility that some choices reflect *mistakes* – failure to make the choice that reflects their preferences – due to the complexity involved in decisions under risk (Martínez-Marquina, Niederle and Vespa, 2019, Oprea, 2024), or in the context of experimentally collected data, the complexity involved in correctly responding to experimental incentives (Plott and Zeiler, 2005, Cason and Plott, 2014, Danz, Vesterlund and Wilson, 2022, de Clippel, Oprea and Rozen, 2024). If choices fail to reflect preferences, the revealed preference argument fails to generate the true welfare-relevant ordering, which poses a threat to subsequent welfare and policy analysis based on the recovered preferences.

A by-product of the mistakes is the stochasticity of choices, i.e., making different choices facing the same decision. If mistakes are stochastic within individuals (Khaw, Li and Woodford, 2021) or heterogeneous across individuals in pooled data sets, choices would become stochastic.[1] To recover preferences from stochastic mistakes, the dominant approach is to first make parametric assumptions over the *stochastic structure* of these choices, i.e., the distribution of choices conditional on the preference. The stochastic structure specifies the possible mistakes that individuals make, along with the probability of each specific mistake. This approach then proceeds by jointly estimating the preferences and the parameters of the stochastic structure. However, the assumptions over the stochastic structure have been shown to greatly affect the recovered preferences (e.g., Buschena and Zilberman, 2000, Carbone and Hey, 2000, Alós-Ferrer, Fehr and Netzer, 2021, O'Donoghue and Somerville, 2024). Making things worse, these assumptions are typically unverifiable, since the stochastic structure can be precisely measured only when the preferences are known, which is typically not the case if one's very goal is to learn these preferences (Alós-Ferrer, Fehr and Netzer, 2021). Therefore, the interpretability of the recovered preferences as a welfare-relevant ordering and applicability in welfare analysis and policy design are severely limited.

[1]We acknowledge that stochastic choices are most commonly interpreted as within-individual fluctuations in *preferences* or unobserved heterogeneity of individual preferences in a pooled data set (e.g., McFadden, 1974). Both interpretations above are preference-based, implying that the stochasticity is conceptually distinct from mistakes and instead, should enter into welfare analysis. As will be made clear later, in this paper, we view stochastic choices as resulting from both mistakes and preferences.

In this paper, we develop an approach that reduces the reliance on unverifiable assumptions about the stochastic structure, and then apply the approach to recover a novel measure of risk preferences. The approach starts from eliciting subjects' valuations of auxiliary objects with a known valuation that is closely related to lotteries – namely, the deterministic mirrors of the lotteries (Oprea, 2024). The deterministic mirrors have two key advantages. First, the mirrors have *objectively correct* valuations, allowing us to directly observe the mistakes and estimate the stochastic structure for mirrors. Second, the mistakes in mirror valuation tasks plausibly serve as a lower bound of those in lottery valuation tasks, in the sense that any mistakes present in the mirror tasks should also manifest in the lottery tasks.

In our experiment, each subject values a set of binary-outcome lotteries ($\$X, p\%; \$0$) (with probability $p\%$ winning a prize of $\$X$, and \$0 otherwise). Each subject also values the deterministic mirror of each lottery that they value. For a lottery ($\$X, p\%; \$0$), the lottery and its mirror are both presented to subjects as 100 boxes, p of which contain $\$X$, with the remaining boxes containing \$0. The key difference is in the payment rule. For lotteries, one box is randomly drawn to determine the payout, creating uncertainty. For mirrors, subjects are told the payout equals the average across all 100 boxes with certainty. This means the mirror pays out the expected value of its corresponding lottery. The valuations of lotteries and mirrors are elicited with price lists (Holt and Laury, 2002) with an enforced single-switching rule, yielding a unique switching point where subjects shift from preferring the lottery or the mirror to preferring a sure payment.

Crucially, both lotteries and mirrors require subjects to process the same disaggregated presentation of outcomes and to navigate the same price list elicitation mechanism. Mistakes arising from the complexity of processing this disaggregated format (Oprea, 2024) or from misunderstanding how price lists work (Andersen et al., 2006, Beauchamp et al., 2020) should therefore manifest similarly in both contexts. Although risk may introduce an additional source of error specific to decision-making under uncertainty (Martínez-Marquina, Niederle and Vespa, 2019), these risk-specific errors operate in addition to, rather than systematically offsetting, the errors shared by lotteries and deterministic mirror tasks. This argument supports using mirror-estimated stochastic structure to correct lottery responses: we can identify and measure the mistakes common to both contexts through mirrors and correct for them in lottery valuations, while recognizing that lotteries may contain additional risk-specific mistakes that stay uncorrected.

Before describing our approach, we need to clarify a few key terms. We use *response* to refer to the switching point the subject actually submits when valuing a lottery or mirror, and *genuine valuation* to refer to the switching point that would lead to the most desirable outcome according to a subject's welfare-relevant preference ordering. When the response differs from the genuine valuation, we refer to this discrepancy as a *mistake*. We define

stochastic structure as the distribution of responses conditional on genuine valuation. In mirror tasks, genuine valuations are known, which, in turn, makes the stochastic structure observable. In lottery tasks, however, genuine valuations are unknown and heterogeneous across individuals, making the stochastic structure unobservable.

Our approach proceeds in two steps to recover a measure of risk preferences referred to as *risk-induced valuations* (RIV). The first step is to estimate the stochastic structure in mirror tasks. We observe that the vast majority of mistakes made by subjects when valuing mirrors belong to either of the two types: *difference-dependent mistakes* or *unconditional mistakes*. Difference-dependent mistakes are errors in which subjects submit valuations that are close to, but not equal to, the genuine valuation, with the probabilities of these responses depending only on the difference between the response and the genuine valuation. Unconditional mistakes are errors in which subjects submit responses drawn from a distribution that is invariant to the genuine valuation. With this observation, we then build a mixture model assuming that the stochastic structure in mirror tasks is a weighted average of three component distributions: the genuine valuation, difference-dependent mistakes, and unconditional mistakes. Estimating this mixture model gives us a complete representation of the stochastic structure that expands to out-of-sample genuine valuations. The unconditional mistakes mainly take the form of submitting the highest or lowest feasible responses, as well as the responses right in the middle of the price list. On the other hand, the difference-dependent mistakes exhibit systematic downward bias for both prizes.

Having estimated the stochastic structure in mirror tasks, the second step applies the estimated stochastic structure from mirror tasks to correct lottery responses and construct RIV. Compared with raw responses, the distributions of the RIV are substantially more concentrated around the risk-neutral benchmarks of the lotteries. When RIVs deviate from risk-neutral benchmarks, they predominantly exhibit risk aversion, with risk seeking being less common. Even for lotteries involving small probabilities of gains ($p = 10$), the RIV distributions show roughly equal proportions of risk-averse and risk-seeking valuations. This contrasts sharply with both the raw responses in our data and the classic fourfold pattern documented by Tversky and Kahneman (1992), where the majority of responses exhibit risk seeking for low-probability gains. Finally, the RIV are less heterogeneous than the raw responses, suggesting that heterogeneity in responses (Bruhin, Fehr-Duda and Epper, 2010) partly reflects heterogeneity in mistakes rather than solely heterogeneity in genuine risk preferences.

These patterns have several implications. First, because the expected utility theory (EUT) predicts approximate risk neutrality under small stakes (Rabin, 2000), the RIV are more aligned with the EUT predictions than the raw responses. The RIV also exhibit near risk neutrality under the smaller prize of \$25 and weak risk aversion under the larger prize of

$100, consistent with EUT under concave utility functions. Together, these patterns suggest that previous literature has overestimated the degree of deviation of risk preferences from EUT. Second, our evidence suggests that past estimates of prospect-theoretic parameters – particularly the strong overweighting of small probabilities – should not be interpreted as solely reflecting genuine preferences, but at least partly reflect heterogeneous mistakes in responding to preference elicitation tasks.

To assess the robustness and generalizability of our findings, we apply the same two-step procedure to two additional data sets: Oprea (2024) and Zhang (2025). Both studies collect valuations of lotteries and mirrors from the same subjects, allowing our approach to be applied seamlessly. The key findings in this paper are robust to using alternative data sets. First, the mixture model with difference-dependent and unconditional mistakes provides an excellent fit to mirror responses in both data sets, demonstrating that these two mistake types are robust features of behavior in mirror valuation tasks across different experimental implementations. Second, the RIV recovered from both data sets exhibit patterns similar to our main results: compared to raw responses, the RIV are more concentrated around risk-neutral benchmarks.

Moreover, we assess our approach using Zhang's (2025) data, which includes the same group of subjects valuing the same set of lotteries and mirrors twice under two different treatments – one with and one without a calculator. The raw-response distributions differ significantly across treatments, as the calculator reduces calculation costs and thus reduces certain types of mistakes. After applying the treatment-specific corrections, however, our joint test does not reject equality of the recovered RIV distributions. This is the direction predicted if the correction removes the treatment-specific, non-risk-specific mistakes while genuine preferences and risk-specific mistakes remain stable. The result therefore passes a necessary-condition check for the maintained interpretation, but non-rejection does not establish equality or equivalence of the RIV distributions.

In terms of the topic, this study contributes to the literature studying the stochasticity of choices under risk (Hey and Orme, 1994, Hey, 1995, 2005, Loomes and Sugden, 1998, Buschena and Zilberman, 2000, Carbone and Hey, 2000, Khaw, Li and Woodford, 2021). Relatedly, there is also a literature studying the roles of non-preference factors in choices under risk (Woodford, 2012, Martínez-Marquina, Niederle and Vespa, 2019, Frydman and Jin, 2021, Nielsen and Rehbeck, 2022, Enke and Graeber, 2023, Barretto-García et al., 2023, Enke and Shubatt, 2023, Oprea, 2024, McGranaghan et al., 2024, Zhang, 2025). Non-preference factors in choices under risk are also found in field settings, especially in the context of insurance choices (e.g., Barseghyan et al., 2013, Handel et al., 2024). This study contributes to these strands of literature by characterizing the stochastic structure of mistakes in mirror tasks through the mixture model and using that structure to correct for mistakes in lottery tasks,

thereby yielding a better measure of genuine risk preferences than raw lottery responses.

In terms of experimental methodologies, this paper follows the lead of Cason and Plott (2014), Martínez-Marquina, Niederle and Vespa (2019), Enke and Shubatt (2023), and Oprea (2024). This literature develops the experimental technique of asking subjects to value auxiliary objects of *known* valuations (e.g., mirrors) that are closely related to the objects of interest (e.g., lotteries), which typically have *unknown* valuations. One focus of this literature has been identifying the structure of mistakes when valuing the auxiliary objects, and utilizing the identified stochastic structure to shed light on the structure of mistakes when valuing the objects of interest. This paper advances this literature in two ways. First, by adopting a mixture modeling approach, we provide a characterization of the stochastic structure when valuing the auxiliary objects. Second, this paper provides a formal way to correct for the identified mistakes when valuing the objects of interest.

In terms of statistical modeling, this paper is related to the literature in psychology that uses mixture modeling to estimate and correct for unconditional mistakes[2] (Wichmann and Hill, 2001*a*,*b*, Treutwein and Strasburger, 1999, Prins, 2012, Clark and Merfeld, 2021). This literature typically applies mixture modeling to binary choice data. The current paper expands the mixture modeling to multinomial choices and even a continuum of possible options. In economics, mixture modeling has been employed to study risk preferences in Bruhin, Fehr-Duda and Epper (2010) and Conte, Hey and Moffatt (2011), and a related hierarchical Bayesian modeling approach is employed in Balcombe and Fraser (2025). While studies in this literature employ these modeling approaches mainly to uncover the heterogeneity of lottery valuations at within-individual or between-individual levels, we rely on mixture modeling to characterize and correct for the mistakes made by subjects.

The rest of this paper is organized as follows. Section 2 describes the experimental design. Section 3 outlines our conceptual framework. Section 4 describes our Step 1 of estimating the stochastic structure in mirror tasks. Section 5 illustrates our Step 2 of recovering risk-induced valuations. Section 6 applies the two-step procedure developed in this paper to the data sets of Oprea (2024) and Zhang (2025). Finally, Section 7 discusses the implications of our findings and concludes.

2 Experimental Design

We elicit subjects' valuations for a set of binary lotteries $(\$X, p\%; \$0)$. Each subject faces five different p in the experiment: $p \in \{10, 25, 50, 75, 90\}$, and is randomized into one of the two prizes \$25 and \$100. The prize stays the same for a fixed subject throughout the experiment. In the experiment, each lottery is presented as 100 boxes, each containing some amount of

[2]The terminology used in the psychology literature for what we refer to as unconditional mistakes is *lapses*.

money. The subjects are told that the realized payout of a lottery will be determined by the amount in a randomly drawn box.

In the same experimental session, subjects are also asked to value the deterministic mirrors of each of the five lotteries (Oprea, 2024). For a lottery, its deterministic mirror is presented visually as the same set of 100 boxes. However, the subjects are told that given a set of boxes, the mirror pays out the average amount of money contained in these 100 boxes with certainty. In other words, a mirror pays out the expected value of its corresponding lottery with certainty.

All valuations are elicited through price lists (Holt and Laury, 2002). An example price list is provided in Table 1. In each row of a price list, the subjects have to indicate whether they would like to choose a lottery/mirror, which is fixed in a price list, or a sure payment, which varies from the top to the bottom of the price list. The chosen option in each row is marked with green shading, as we do in the actual experimental interface. Each price list has 26 rows with equal increments – under prize \$25, subjects choose between a lottery/mirror and 26 sure payments ranging from \$0 to \$25, with increments of \$1; under prize \$100, subjects choose between a lottery/mirror and 26 sure payments ranging from \$0 to \$100, with increments of \$4.

Decision	**Option A**	**Option B**
1	Being paid according to the boxes	**\$0.00** for sure
2	Being paid according to the boxes	**\$1.00** for sure
3	Being paid according to the boxes	**\$2.00** for sure
4	Being paid according to the boxes	**\$3.00** for sure
5	Being paid according to the boxes	**\$4.00** for sure
⋮	⋮	⋮
25	Being paid according to the boxes	**\$24.00** for sure
26	Being paid according to the boxes	**\$25.00** for sure

Table 1: An example price list

We enforce single switching in the price lists: if a subject chooses a sure payment $\$x$ over the lottery/mirror, they have to also choose $\$x'$ over the lottery/mirror for all $x' > x$. Due to enforced single switching, for each price list, there is a highest sure payment $\$x^*$ that the subject chooses the lottery/mirror over $\$x^*$, but chooses $\$x'$ over the lottery/mirror for every $x' > x^*$.[3] We refer to this highest x^* as the *response*. For example, in Table 1, the response is \$2.[4]

[3]Two special cases are: (1) the subject always chooses the sure payment; and (2) the subject always chooses the lottery/mirror. In the former case, the switching point is coded as \$0, while in the latter, the switching point is coded as the prize.

[4]The response differs slightly from the certainty equivalent, since the latter is the sure payment that

The experiment is split into two blocks – one containing all lottery tasks and the other containing all mirror tasks. The order of blocks is randomized at the subject level. The subjects are not informed about the second block while completing the first block, but receive instructions about the new block before starting the second block. Before each block, the subjects are first explained the payment rule (lottery or mirror) in the coming block. Then, the subjects are shown three examples. In each example, we present them with a set of 100 boxes, and generate 10 simulated payouts from the set of boxes according to the payment rule that is currently in effect. Finally, the subjects are asked to answer the five comprehension questions from Wu (2025) that test their understanding of the payment rule in effect. If a subject fails two or more comprehension questions in any of the two blocks, they will be screened out of the experiment. The subjects only start a block after passing the comprehension check. The instructions for the price lists are adopted from McGranaghan et al. (2024). The complete experimental instructions are provided in the Appendix.

The experiment was conducted on Prolific in May 2025. Each subject received a $3 payment upon completion. We incentivized the responses using the following mechanism: At the end of the experiment, with a probability $1/X$, where X is the subject-specific prize,[5] a row in a price list was randomly selected, and the subject's choice in that row determined their payment. The median completion time was 22 minutes, and the average total payment was $3.51. In total, 292 subjects finished the experiment, of whom 149 were assigned to prize $X = 25$ and 143 to prize $X = 100$. In addition, 214 subjects failed the comprehension check in either of the blocks and were screened out.[6] Our data include only the subjects who finished the experiment.

3 Conceptual Framework

Let $\mathcal{L}$ denote the set of lotteries and let $\mathcal{R}$ denote the finite set of feasible responses. Consider a randomly drawn subject facing lottery $l \in \mathcal{L}$. Let the random variable $R_l \in \mathcal{R}$ denote the subject's observed response. Let the random variable $V_l \in \mathcal{R}$ denote the response the subject would submit under mistake-free evaluation and implementation. We call V_l the subject's *genuine valuation*. Genuine valuations may vary across the subject population. For each lottery l, define

$$q_l(v) \equiv \Pr(V_l = v), \qquad v \in \mathcal{R}.$$

makes a subject *indifferent* between the lottery and the sure payment. In our design with discrete payment increments, the certainty equivalent lies between the response and the next higher sure payment.

[5]This payout probability is set to be consistent with the design of Wu (2025).

[6]If a subject passed the comprehension check of the first block but failed the second, we paid them $1.5 to compensate for their time.

Thus, q_l is the population distribution of genuine valuation V_l for lottery l. All probabilities in this section refer to the pooled population distribution of a randomly drawn subject; our analysis is distributional rather than individual-level.

Subjects may submit responses that differ from their genuine valuations because of inattention, misunderstanding of the price list, calculation errors, simplifying heuristics, perceptual noise, or other non-preference factors. We refer to a discrepancy between R_l and V_l as a *mistake*. Define the conditional lottery-response distribution for each lottery l

$$g_l(r \mid v) \equiv \Pr(R_l = r \mid V_l = v), \qquad r, v \in \mathcal{R},$$

which we call the *stochastic structure* in lottery tasks. It summarizes the probabilistic relationship between what subjects genuinely prefer and what they report.

Let

$$p_l(r) \equiv \Pr(R_l = r)$$

denote the marginal distribution of observed responses for lottery l. By the law of total probability,

$$p_l(r) = \sum_{v \in \mathcal{R}} g_l(r \mid v) q_l(v). \tag{1}$$

Lottery valuation data identify p_l, but Equation (1) reveals a fundamental identification problem: g_l and q_l are not separately identified. Without restrictions on the stochastic structure, the distribution of genuine valuations q_l cannot be recovered from observed lottery responses alone.

Facing this identification problem, the traditional approach is to impose a parametric stochastic structure. For example, the *Normal Mistake* model assumes

$$g_l(r \mid v) \propto \exp\left(-\frac{(r-v)^2}{\sigma^2}\right), \qquad \text{for some } \sigma > 0.$$

With such a restriction, one can jointly estimate q_l and the parameters of g_l, such as σ. If the assumed functional form is incorrect, however, the recovered q_l will generally be biased. For example, experimental evidence indicates that multiple price lists exhibit a pull-to-the-center effect that can shift responses toward the center of the list (Andersen et al., 2006, Beauchamp et al., 2020). A Normal Mistake model is incompatible with this specific stochastic structure, and would in turn cause center-seeking mistakes to be misinterpreted as concentration of genuine valuations near the center.

Deterministic mirrors are an approach to tackle this identification problem because they reveal how subjects translate a known value into a reported response. A deterministic mirror

has an objectively correct response $v \in \mathcal{R}$, known by design. Write

$$g_M(r \mid v) \equiv \Pr(\text{report } r \mid \text{correct mirror response } v)$$

for the probability of reporting r in a deterministic mirror whose correct response is v. Because the mirror contains no risk and has a known correct value, deviations from v reveal mirror-measured distributions of non-risk-specific mistakes.

To connect the stochastic structures in lotteries and mirrors, we observe that when subjects submit their responses, two types of mistakes can happen: risk-specific and non-risk-specific. Martínez-Marquina, Niederle and Vespa (2019) document failure of contingency reasoning in the presence of risk, which is a risk-specific mistake. Meanwhile, the pull-to-the-center effect of price lists is an example of a non-risk-specific mistake. If both types of mistakes are present, correcting either type would usually result in better measurement of risk preferences than the distribution of raw responses. Later we provide precise conditions under which this is true.

We hypothesize that subjects generate a response in a lottery valuation task through a sequential representation process. In the first stage, subjects make risk-specific mistakes and map their genuine valuation V_l into an intermediate value, referred to as *risk-induced valuation* (RIV) and denoted as W_l. In the second stage, subjects make non-risk-specific mistakes that map the intermediate value W_l to a response R_l.

Because each deterministic mirror has a known correct value, deviations of reported values from it reveal mistakes that arise in the absence of risk. The mirror and lottery valuation tasks use closely matched monetary amounts, visual displays, response options, and elicitation procedures. These design similarities make it plausible that the same non-risk-specific mistakes operate in both tasks. For example, evidence that similar mental representations of numerical magnitudes are used in perceptual and risky-choice tasks suggests that imprecision in numerical perception may affect responses in both settings (Barretto-García et al., 2023). Another example is, if a subject literally trembles his hand when trying to submit a response, this trembling hand error should appear in lottery and mirror tasks with the same frequency. We therefore treat the sharing of the mirror-measured non-risk-specific mistakes across the two tasks as a maintained assumption.

To simplify notation, let $\boldsymbol{p_l}$ and $\boldsymbol{q_l}$ collect $p_l(r)$ and $q_l(v)$ as column vectors. Let G_l and G_M be the stochastic matrices with entries $g_l(r \mid v)$ and $g_M(r \mid v)$, respectively. Formally, we make the following assumption.

Assumption 1. We assume that there exists a risk-specific mistake structure $g_l^{RS}(w \mid v) \equiv \Pr(W_l = w \mid V_l = v)$ for each l such that the stochastic structures in lottery and mirrors have

the following relationship

$$g_l(r \mid v) = \sum_{w \in \mathcal{R}} g_M(r \mid w) g_l^{RS}(w \mid v) \quad \forall r, v \in \mathcal{R}.$$

In the vectorized form,

$$G_l = G_M G_l^{RS} \quad \text{for all } l \in \mathcal{L}$$

where G_l^{RS} is a stochastic matrix of risk-specific mistakes with entries $g_l^{RS}(w \mid v)$.

Assumption 1 captures the hypothesized sequential representation through which subjects respond to lottery valuation tasks. Under the vectorized notation, Equation (1) can be rewritten as

$$\boldsymbol{p_l} = G_l \mathbf{q_l}. \tag{2}$$

Equation (2) can be broken down into the two sequential representation stages. The first stage – risk-specific mistakes G_l^{RS} – maps V_l to W_l and induces a distribution of RIV for lottery l, denoted $\boldsymbol{w_l} \equiv (\Pr(W_l = w))_{w \in \mathcal{R}}$, according to

$$\boldsymbol{w_l} = G_l^{RS} \boldsymbol{q_l},$$

and the second stage – non-risk-specific mistakes G_M – that maps W_l to R_l further induces the distribution of responses

$$\boldsymbol{p_l} = G_M \boldsymbol{w_l} = G_M G_l^{RS} \boldsymbol{q_l} \equiv G_l \boldsymbol{q_l}.$$

In light of this hypothesized sequential representation process, we propose a two-step procedure that aims to correct for the non-risk-specific mistakes that can be identified by the deterministic mirrors, while leaving the unidentified risk-specific mistakes in the first stage uncorrected. To be more specific, in the first step of our procedure, we estimate the stochastic structure G_M in mirror tasks. This is relatively straightforward because mirrors have objectively correct responses and the response space $\mathcal{R}$ is finite. The estimate is denoted $\hat{G}_M$. Then, in the second step of our procedure, we could ideally premultiply the observed response distribution $\hat{\boldsymbol{p}}_l$ by the inverse of $\hat{G}_M$ to estimate the RIV distribution:

$$\hat{\boldsymbol{w}}_{\boldsymbol{l}} = \hat{G}_M^{-1} \hat{\boldsymbol{p}}_l. \tag{3}$$

This $\hat{\boldsymbol{w}}_{\boldsymbol{l}}$ will be the final output of our procedure.

Under Assumption 1, Equation (3) yields an estimate of the distribution of W_l. It retains both genuine preference heterogeneity and all lottery-specific mistakes. It therefore need not equal the distribution of genuine valuations. Assumption 1 alone implies neither that W_l is

closer to V_l than R_l, nor that W_l is close to V_l in an absolute sense. We will get back to these distinct questions after constructing the RIV distribution from our data.

4 The Stochastic Structure in Mirror Tasks

4.1 Distributions of Responses

Figure 1 shows the distributions of responses for each deterministic mirror. The objectively correct valuation is indicated by the red vertical line. We highlight several consistent features of these distributions.

Obs 1 The peak of the distribution is always the genuine valuation. For a given prize, the probability mass at the genuine valuation is *remarkably stable* across p: around 0.3 for prize \$25 and around 0.5 for prize \$100.

Obs 2 Fixing a prize, the probability masses of responses adjacent to the peak are *stable* across p. For example, when the prize is \$25, the responses immediately before the peak have probability masses around 0.15.

Obs 3 Regardless of p, there are non-negligible probability masses at the extreme responses (e.g., \$0, \$24, \$25 under prize \$25), and at certain mid-range responses (e.g., \$9 and \$10 under prize \$25). The probability masses at these responses are *stable*.[7]

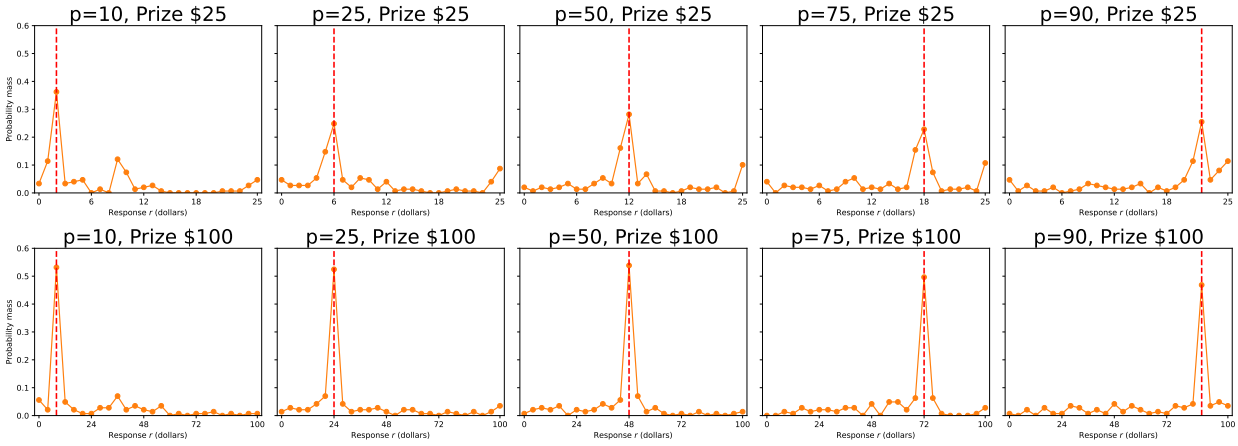


Figure 1: Distributions of responses in mirror tasks

These observations suggest that the responses in mirror tasks mainly fall into three types, one rational choice and two types of mistakes. First, some subjects perfectly understand

[7]This observation is stronger under prize \$25, but is still visible under prize \$100.

the task and choose the genuine valuation (Obs 1). Second, instead of choosing the genuine valuation, some subjects commit *difference-dependent mistakes* by choosing a response close to the rational one (Obs 2). Such mistakes likely arise from subjects who generally understand the concept of mirrors and respond systematically to the probability of the corresponding lottery, but who occasionally miscalculate the mirror's payout or misreport their choice in the price lists. Finally, some subjects commit *unconditional mistakes* by completely ignoring changes in the probabilities (Obs 3). This type of mistake potentially reflects subjects who are inattentive throughout the experiment, or who employ some decision-making heuristic that disregards information about p. Moreover, the relative frequencies of these three types of responses are stable across mirror tasks fixing a prize, as indicated by the stability of the probability masses documented in Observations 1–3.

Crucially, the stochastic structure we observe in mirror tasks is inconsistent with commonly employed assumptions about the stochastic structures in valuation tasks. For example, Bruhin, Fehr-Duda and Epper (2010) assume that the stochastic structure adds a zero-mean normal noise to the genuine valuation, which would imply the distribution of responses to be normally distributed and symmetric around the genuine valuation. Meanwhile, Barseghyan et al. (2013) assume, following McFadden (1974), that the noise is a Type-I extreme value distribution added to the utility. This assumption would lead to a single-peaked probability mass function of responses, with the peak at the genuine valuation. Neither assumption is consistent with the actual stochastic structure in Figure 1. As a result, had we not known the genuine valuations in mirrors and needed to estimate them from the responses, adopting these assumptions would lead to inaccurate conclusions about the genuine valuations.

The three types of responses identified above, together with the stability of their frequencies, naturally motivate a model that represents the distribution of responses as a probability mixture of three component distributions, each corresponding to one type of responses. This framework is commonly referred to as *mixture models*, and it turns out to be immensely useful for estimating G_M.

4.2 Mixture Modeling of Responses

We now separately estimate G_M for the two prizes \$25 and \$100 as the first step of our procedure. G_M is different for the two prizes for a few reasons. First, we use different price lists for each prize, which leads to different $\mathcal{R}$ and different mistakes. Second, the two prizes \$25 and \$100 naturally involve different levels of computational difficulty if one wants to calculate the correct values of the mirrors.

We introduce some notation. In our actual experiment, under a fixed prize, we do not elicit valuations of all possible mirrors, but only a subset of mirrors $\mathcal{M}$. For each mirror m, the genuine valuation is known and denoted $v(m)$. The set of *in-sample* genuine valuations

is $V_{\mathcal{M}} = \{v(m) : m \in \mathcal{M}\} \subseteq \mathcal{R}$, while the set of *out-of-sample* genuine valuations is $V_{\mathcal{M}}^c = \mathcal{R} \setminus V_{\mathcal{M}}$. To make the definitions concrete, under prize \$25, the set of possible responses is $\mathcal{R} = \{0, 1, 2, \ldots, 25\}$, and the set of in-sample genuine valuations is $V_{\mathcal{M}} = \{2, 6, 12, 18, 22\}$. Under prize \$100, the set of possible responses is $\mathcal{R} = \{0, 4, 8, \ldots, 100\}$, and the set of in-sample genuine valuations is $V_{\mathcal{M}} = \{8, 24, 48, 72, 88\}$.[8]

Before describing the mixture model in detail, we would like to explain why we need a model for $g_M(r|v)$ at all. Our immediate goal is to estimate the complete stochastic matrix $G_M \equiv \{g_M(r|v)\}_{r,v\in\mathcal{R}}$. For in-sample genuine valuations ($v \in V_{\mathcal{M}}$), the conditional probabilities $\{g_M(r|v) : r \in \mathcal{R}, v \in V_{\mathcal{M}}\}$ can be estimated directly by the empirical frequencies of r conditional on v, as illustrated in Figure 1. However, for out-of-sample genuine valuations ($v \in V_{\mathcal{M}}^c$), without further assumptions, we do not know the conditional probabilities $\{g_M(r|v) : r \in \mathcal{R}, v \in V_{\mathcal{M}}^c\}$. To expand the domain of $g_M(r|v)$ from in-sample to out-of-sample, we can exploit the empirical regularities documented in Observations 1–3 by specifying a statistical model that incorporates the three observed types of responses, and use the model to *predict* $g_M(r|v)$ for out-of-sample v. The mixture model serves exactly this purpose, and as we show later, it indeed achieves strong in-sample fit and out-of-sample predictive power.

Formally, the mixture model is specified in Model 1 that describes $g_M(r|v)$ as a probability mixture of three component distributions:

1. A degenerate *rational distribution* $\mathbb{1}[r = v]$, which assigns probability one on the genuine valuation v;

2. A *difference-dependent* mistake distribution $h_{dd}\left(\frac{r-v}{S}\right)$, which allocates probability based only on the number of price list rows between r and v: $\frac{r-v}{S}$, where S is the step size between two adjacent rows. We restrict the support of difference-dependent mistakes to within K rows.

3. An *unconditional* mistake distribution $h_{uc}(r)$, which allocates probability mass independent of v.

[8]For example, for the mirror of the lottery (\$100, 10%; \$0), which involves a sure payout of \$10, the subject should choose a response of \$8, since the subject prefers the mirror to \$8, but prefers \$12 (the sure payment in the next row) to the mirror. For the other mirrors, their corresponding $v(m)$ are computed accordingly.

Model 1 (Mixture Model for Mirror Responses)**.**

$$g_M(r|v;\Psi) = w_{rational}\mathbb{1}[r=v] + w_{dd}h_{dd}\left(\frac{r-v}{S}\right) + w_{uc}h_{uc}(r) \tag{4}$$

$$s.t \quad w_{rational} + w_{dd} + w_{uc} = 1$$

$$h_{dd}(x) = \begin{cases} \pi_x & \text{if } x \in \{-K,\ldots,-1,1,\ldots,K\} \\ 0 & \text{otherwise} \end{cases} \tag{5}$$

$$s.t \quad \sum_{x\in\{-K,\ldots,-1,1,\ldots,K\}} \pi_x = 1$$

$$h_{uc}(r) = \lambda_r \tag{6}$$

$$s.t \quad \sum_{r\in\mathcal{R}} \lambda_r = 1$$

The complete set of parameters in this model is

$$\Psi = \{w_{rational}, w_{dd}, w_{uc}, \{\pi_x\}_{x\in\{-K,\ldots,-1,1,\ldots,K\}}, \{\lambda_r\}_{r\in\mathcal{R}}\}.$$

Crucially, since the mixture model is set up and estimated under flexible functional forms, we avoid any bias from misspecified functional forms in the process of domain expansion. The difference-dependent mistake distribution $h_{dd}\left(\frac{r-v}{S}\right)$ is parameterized by probabilities $\{\pi_x\}$ that sum up to one.[9] The unconditional distribution $h_{uc}(r)$ is parameterized by probabilities $\{\lambda_r\}_{r\in\mathcal{R}}$, also summing to one. In the model, the window size of the difference-dependent mistake distribution, K, is a "hyperparameter," which is not estimated, but chosen by the econometrician. In our main specification, we choose $K = 2$. Other choices of K yield similar results.

Let $D_M = \{(i, m, r_{m,i})\}_{i,m}$ denote the indexed mirror-valuation sample, where $r_{m,i} \in \mathcal{R}$ is subject i's response to mirror $m \in \mathcal{M}$. The log likelihood function of this model is given by

$$\mathcal{L}_M(\Psi; D_M) = \sum_{(i,m,r_{m,i})\in D_M} \log g_M(r_{m,i}|v(m);\Psi). \tag{7}$$

We estimate the parameters Ψ by maximum likelihood estimation

$$\hat{\Psi} \in \arg\max_{\Psi} \mathcal{L}_M(\Psi; D_M),$$

[9]For corner cases $v < KS$ or $v > \bar{r} - KS$ ($\bar{r} \equiv \max_{r\in\mathcal{R}} r$), Equation (5) would assign probabilities to out-of-range $r < 0$ or $r > \bar{r}$. To restore the constraint that probabilities sum up to one, when implementing Model 1, we shift all probabilities assigned to the out-of-range $r < 0$ to 0, and $r > \bar{r}$ to $\bar{r}$. Other approaches to deal with these corner cases yield similar results over the RIV.

using the expectation-maximization (EM) algorithm (Dempster, Laird and Rubin, 1977), which is a standard method for estimating mixture models.

4.3 The Estimated Mixture Model

Figure 2 presents the estimated distributions of the two mistake components, h_{dd} and h_{uc}, under prize \$25. The difference-dependent mistakes occur primarily when subjects choose the response immediately below the genuine valuation, while other deviations are relatively rare. The unconditional mistakes disproportionately place mass on the highest response (25), as well as on mid-range responses (9 and 10) and the lowest response (0). In terms of estimated mixture weights, unconditional mistakes account for $w_{uc} = 0.538$, which is more than twice the weight of the difference-dependent mistakes ($w_{dd} = 0.203$). Both types of mistakes are important, however, with magnitudes comparable to the rational component ($w_{rational} = 0.259$).

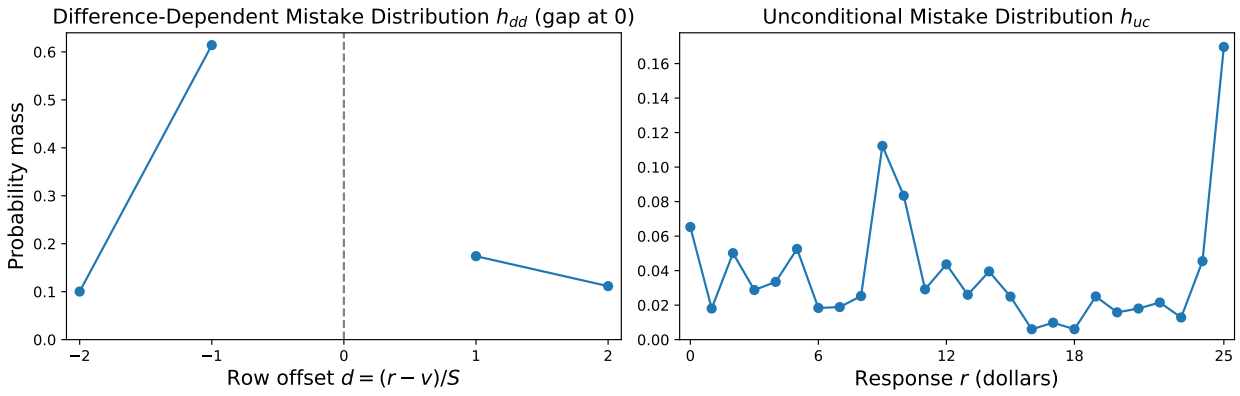


Figure 2: The distributions h_{dd} and h_{uc} under prize \$25

Note: h_{dd}'s domain is $\{-2, -1, 1, 2\}$ due to our choice of $K = 2$.

Figure 3 reports the corresponding estimates under prize \$100. Difference-dependent mistakes are concentrated on responses just below or just above the genuine valuation, but not further away. Unconditional mistakes are most prominent at the mid-range points (e.g., 36, 48, 56), the low responses (e.g., 8, 16), and the two highest responses (96 and 100). The estimated mixture weights again indicate that unconditional mistakes are substantially more frequent than difference-dependent mistakes ($w_{uc} = 0.396$ vs. $w_{dd} = 0.108$).

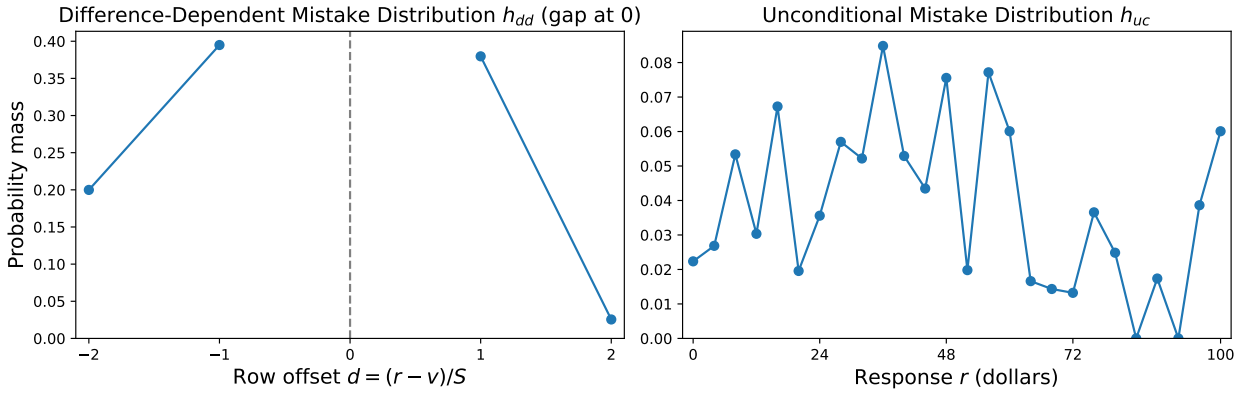


Figure 3: The distributions h_{dd} and h_{uc} under prize \$100

Note: h_{dd}'s domain is $\{-2, -1, 1, 2\}$ due to our choice of $K = 2$.

To assess overall fit, Figure 4 juxtaposes the empirical distributions of responses with the distributions simulated from the estimated mixture model. The simulated distributions reproduce the key empirical features documented in Observations 1–3, delivering a great in-sample fit. Importantly, the good fit does not arise simply from flexibility of the functional forms of h_{dd} and h_{uc}. Despite the flexibility, the mixture model imposes strong restrictions on the distributions of the responses by requiring the distributions to be fixed-weight mixtures of the three types of responses. In Appendix A, we further show that the mixture model has not just great in-sample fit, but strong out-of-sample predictive power, which is not achievable through functional form flexibility alone. The close alignment between simulated and empirical distributions therefore provides strong evidence in favor of the three-type interpretation of the mirror response distributions.

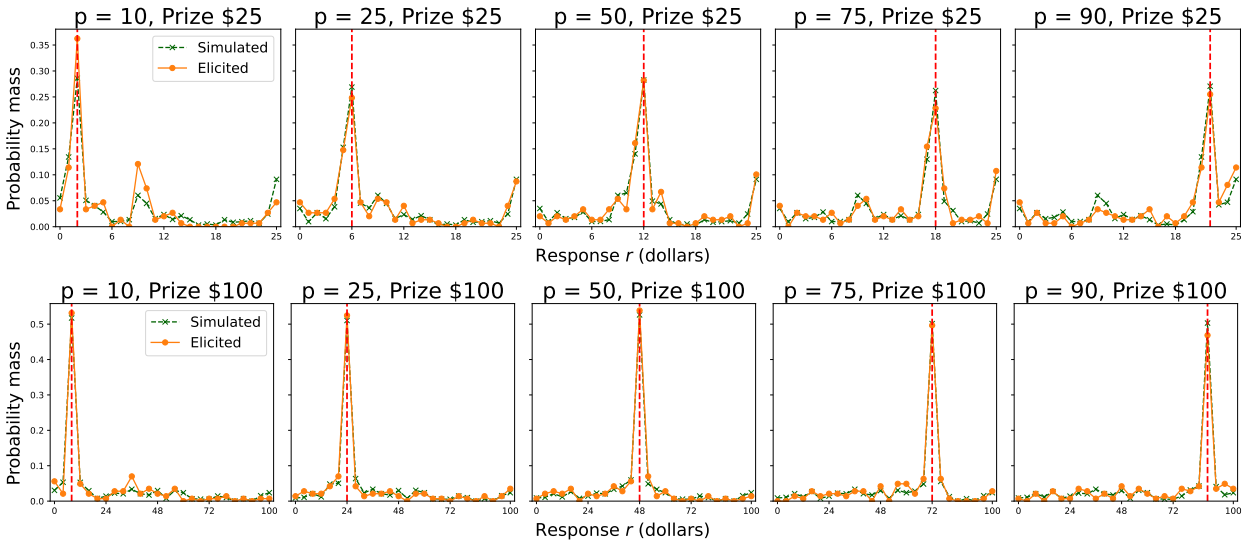


Figure 4: Simulated distribution of mirror responses against the empirical distribution

5 Estimating Risk-Induced Valuations

Having estimated the stochastic structure in mirror tasks in Section 4, we now apply it to lottery responses to recover risk-induced valuations. Lottery tasks differ from mirror tasks in two important respects. First, unlike in mirror tasks, the genuine valuation is *unknown*. Second, the genuine valuation may be *heterogeneous* across subjects.

5.1 Distributions of Responses

Figure 5 shows the distributions of responses in lottery tasks. The risk-neutral benchmark – the response that would have been chosen by a risk-neutral agent who makes no mistakes – is marked with a red vertical line. Three observations emerge that are parallel to Observations 1-3 in mirror tasks.

Obs 1' Unlike the mirror tasks, where the distribution features a sharp single peak at the genuine valuation, the lottery tasks exhibit a *peak area* – a cluster of responses with relatively high probability mass. This peak area is typically centered around the risk-neutral benchmark of the lottery.

Obs 2' For a given prize, the height of the peak area is approximately stable across different probabilities p.

Obs 3' As in the mirror tasks, there are substantial and stable probability masses at the extreme responses and at selected mid-range responses.



Figure 5: Distributions of responses in lottery tasks

Together, these observations suggest that the empirical regularities documented in the mirror tasks – the existence of peaks, stable near-peak deviations, and stable extreme or

mid-range probability masses – seem to carry over to the lottery tasks in modified form. The key difference is that the heterogeneity of genuine valuations in lottery tasks makes the responses less concentrated. Similar response patterns – with stable fractions of extreme or near-peak responses – also appear in other lottery valuation data sets (e.g., Enke and Graeber, 2023).

5.2 Estimating the RIV Distributions

Having estimated the mirror stochastic structure $\hat{G}_M$ in Section 4, we implement the second step of the procedure in Equation (3) separately for each lottery. Directly applying the sample analogue $\hat{G}_M^{-1}\hat{\boldsymbol{p}}_l$ may produce a vector outside the probability simplex. We therefore estimate the RIV distributions by constrained maximum likelihood.

For each lottery l, write

$$w_{l,x} \equiv \Pr(W_l = x), \qquad \boldsymbol{w}_l \equiv (w_{l,x})_{x\in\mathcal{R}}\,.$$

Here, x denotes a possible RIV, while $w_{l,x}$ is the probability assigned to that value by $\boldsymbol{w}_l$. Thus, a valid $\boldsymbol{w}_l$ satisfies

$$w_{l,x} \geq 0 \quad \text{for every } x \in \mathcal{R}, \qquad \sum_{x\in\mathcal{R}} w_{l,x} = 1.$$

For each lottery l, let $r_{l,i}$ denote subject i's response. Conditional on $\hat{G}_M$, we estimate the distribution of RIV through constrained maximum likelihood as

$$\hat{\boldsymbol{w}}_l \in \underset{\boldsymbol{w}_l\in\Delta(\mathcal{R})}{\arg\max} \sum_i \log\left[\sum_{x\in\mathcal{R}} \hat{g}_M(r_{l,i} \mid x) w_{l,x}\right], \tag{8}$$

where the outer summation is over the subjects who face lottery l. We obtain the maximizer using the EM algorithm. If $\hat{G}_M^{-1}\hat{\boldsymbol{p}}_l$ is a valid probability distribution, it also solves the constrained maximization problem above. Otherwise, the constraints ensure that $\hat{\boldsymbol{w}}_l$ remains a valid probability distribution.

5.3 The Risk-Induced Valuations

Figure 6 shows the distributions of RIV along with the empirical distributions of responses. Table 2 provides the averages and medians of the distributions of RIV, along with the 95% confidence intervals from the subject-clustered whole-pipeline bootstrap.

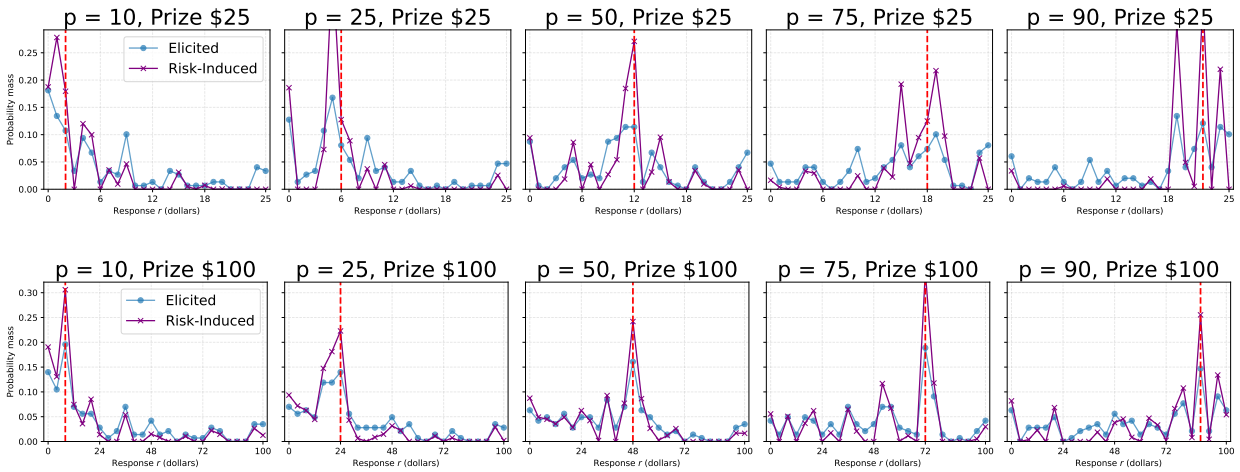


Figure 6: Distributions of risk-induced valuations

Comparing RIV with responses, the most striking observation is that RIV is generally closer to the risk-neutral benchmark than responses. This distinction is evident in both the complete distributions in Figure 6 and the summary statistics in Table 2. As for the distributions, the RIV are more concentrated around the risk-neutral benchmark than the responses, while there are fewer extreme RIV than extreme responses.[10] More quantitatively, we measure the fraction of responses that fall within the window consisting of the risk-neutral benchmark and the two responses immediately above it and the two immediately below it. Under prize \$25, 64.2% of RIV are within this window, while only 42.0% of responses are. Under prize \$100, the numbers are 55.1% and 41.7%, respectively. Turning to the summary statistics for the RIV, under prize \$25, the median RIV are fairly close to the risk-neutral benchmark, as well as the average RIV. Under prize \$100, both the median and the average RIV exhibit weak risk aversion[11] in terms of RIV, except for $p = 10$ where the average is risk-seeking and the median is risk-neutral. In contrast, under both prize levels and regardless of the choice of the summary statistic, the raw responses exhibit pronounced risk seeking when facing a low-probability prize and pronounced risk aversion when facing a high-probability prize, replicating the "fourfold pattern" documented in Tversky and Kahneman (1992).

[10]By extreme responses, we refer to those responses that would manifest extremely risk-averse or risk-seeking preferences, such as choosing a response of \$24 facing the lottery $(\$25, 10\%; \$0)$ or a response of \$0 facing the lottery $(\$25, 90\%; \$0)$.

[11]We use *risk aversion* to refer to the scenario where the RIV is smaller than the risk-neutral benchmark, but not in a strict welfare-relevant sense due to the above-mentioned caveats of interpreting RIV as genuine valuations. The terminology is used similarly for other risk attitudes.

Panel A: Prize $25

p (%)	Risk-neutral	Mean		Median		Share of RIV		
	Benchmark	RIV	Responses	RIV	Responses	Risk-averse	Risk-neutral	Risk-seeking
10	2	3.04 [2.44,5.10]	6.44 [5.32,7.62]	2 [1.00,4.00]	4 [2.00,5.00]	0.47	0.18	0.35
25	6	5.26 [4.35,6.99]	8.10 [6.99,9.24]	5 [5.00,6.00]	6 [5.00,7.00]	0.67	0.13	0.20
50	12	10.66 [9.31,12.39]	11.35 [10.25,12.42]	11 [10.00,12.00]	11 [10.00,12.00]	0.51	0.27	0.22
75	18	16.30 [14.32,17.53]	14.44 [13.28,15.54]	17 [15.00,19.00]	15 [14.00,17.00]	0.50	0.13	0.37
90	22	20.28 [18.02,21.17]	16.80 [15.49,17.99]	22 [19.00,22.00]	19 [19.00,21.00]	0.43	0.35	0.22

Panel B: Prize $100

p (%)	Risk-neutral	Mean		Median		Share of RIV		
	Benchmark	RIV	Responses	RIV	Responses	Risk-averse	Risk-neutral	Risk-seeking
10	8	16.89 [12.81,25.15]	26.52 [21.90,31.19]	8 [8.00,12.00]	12 [8.00,20.00]	0.32	0.31	0.37
25	24	22.02 [19.13,29.09]	29.85 [25.87,33.93]	20 [16.00,24.00]	24 [20.00,24.00]	0.60	0.22	0.18
50	48	35.38 [30.91,42.24]	38.63 [34.60,42.49]	44 [28.00,48.00]	40 [36.00,44.00]	0.57	0.24	0.19
75	72	53.62 [47.51,59.71]	50.49 [46.07,54.88]	64 [52.00,72.00]	52 [48.00,60.00]	0.50	0.34	0.15
90	88	68.00 [60.05,73.53]	60.76 [55.38,66.07]	80 [72.00,88.00]	68 [56.00,80.00]	0.55	0.26	0.19

Table 2: Summary statistics of risk-induced valuations and responses.

Notes: Mean and median cells report point estimates above percentile 95% confidence intervals. Intervals use $B = 2{,}000$ subject-clustered bootstrap replications. Each whole-pipeline RIV replication re-estimates both Model 1 and the RIV distribution; response summaries are recomputed under the same subject-level resampling design.

The shift from responses to risk-induced valuations in Figure 6 and Table 2 can be understood through the stochastic structure in mirror tasks. Recall from Section 4 that subjects make unconditional mistakes by placing weight on extreme or mid-range responses, regardless of p. The mixture model uses this empirical pattern of mistakes to reweight the lottery data. In particular, an extreme or mid-range response observed in a lottery task is classified as more likely to arise from unconditional mistakes unrelated to risk than from the underlying risk-induced valuations. By contrast, a response that is not flagged as a typical mistake in the mirror tasks is more likely to be attributed to risk. In this way, the model

systematically discounts probability mass at extreme and mid-range responses.

Moreover, Table 2 lists the fractions of risk-averse, risk-neutral, and risk-seeking RIV. The majority of deviations from the risk-neutral benchmark are in the direction of risk aversion, and risk seeking is less common. Even for the lotteries involving a prize with probability $p = 10$, risk-averse RIV are as common as risk-seeking RIV. This pattern stands in clear contrast with the distributions of the responses, where under $p = 10$, the majority of responses are risk-seeking,[12] as documented in Tversky and Kahneman (1992) and replicated in countless follow-up works.

The two findings above – concentration of RIV around the risk-neutral benchmarks and predominant risk aversion among deviations from those benchmarks – have a few implications. First, since the expected utility theory (EUT) predicts approximate risk neutrality under small stakes (Rabin, 2000), the RIV are more in line with the predictions of EUT than the responses. In addition, the fact that the RIV distributions exhibit near risk neutrality under the smaller prize of \$25 and weak risk aversion under the larger prize of \$100 is consistent with EUT under concave utility functions. Together, these patterns suggest that the previous behavioral economics literature has overestimated the degree of deviation of risk *preferences* from EUT. Second and relatedly, our evidence suggests that the past estimates of prospect-theoretic parameters in the literature should not be interpreted as solely reflecting genuine preferences. In particular, the strong overweighting of small probabilities documented in the literature should not be solely attributed to genuine preferences for gambling, but at least partly to mistakes in responding to preference elicitation tasks. Finally, the RIV are less heterogeneous than the responses, suggesting the heterogeneity of *responses* in the population (e.g., Bruhin, Fehr-Duda and Epper, 2010) exaggerates the underlying heterogeneity of risk preferences.

5.4 Relative Accuracy: Is the RIV Closer to Genuine Valuations Than Raw Responses?

Section 3 characterizes what the RIV represents, but it does not by itself imply that the RIV is more accurate than the raw response. To make this comparison, let $\ell(a, v) \geq 0$ denote the loss function from using a as a measure of genuine valuation v, with the loss minimized at $a = v$. For lottery l, define

$$\Delta_l \equiv \mathbb{E}[\ell(R_l, V_l)] - \mathbb{E}[\ell(W_l, V_l)]\,.$$

The RIV-stage valuation W_l is weakly more accurate than the raw response R_l under loss ℓ if and only if $\Delta_l \geq 0$.

[12] This can be seen from the medians of the responses.

For this comparison, we additionally maintain that Model 1 captures the actual non-risk-specific mistakes in lottery tasks: within each prize condition, its mixture weights and its two mistake components govern the final mapping from W_l to R_l. Let R_l^{dd} denote the response generated from by a difference-dependent mistake, and let $U \sim h_{uc}$ denote an unconditional-mistake response. Model 1 then represents the observed response as

$$R_l = \begin{cases} W_l, & \text{with probability } w_{\text{rational}}, \\ R_l^{dd}, & \text{with probability } w_{dd}, \\ U, & \text{with probability } w_{uc}. \end{cases}$$

Because the rational component leaves W_l unchanged and the three mixture weights sum to one, the relative loss decomposes exactly as

$$\Delta_l = w_{dd}\Gamma_{dd,l} + w_{uc}\Gamma_{uc,l},$$

where

$$\Gamma_{dd,l} \equiv \mathbb{E}\Big[\ell(R_l^{dd}, V_l) - \ell(W_l, V_l)\Big],$$
$$\Gamma_{uc,l} \equiv \mathbb{E}[\ell(U, V_l) - \ell(W_l, V_l)].$$

Accordingly, RIV is weakly more accurate than the raw response when $\Delta_l \geq 0$.

These two terms capture the forces determining whether RIV is more accurate than raw responses. The unconditional component $\Gamma_{uc,l}$ is positive if a response generated by the unconditional mistakes is, on average, less accurate than the RIV. This restriction is behaviorally plausible, and we maintain $\Gamma_{uc,l} > 0$ as an assumption, though we note that it is not identified by our data because genuine valuations are unobserved. On the other hand, the difference-dependent component $\Gamma_{dd,l}$ can be negative when difference-dependent mistakes correct risk-specific mistakes already contained in W_l. Since our estimates of difference-dependent mistakes exhibit a downward bias, correcting for these mistakes may turn out to increase the loss when risk-specific mistakes exhibit an upward bias. The exact condition for the RIV to have weakly lower expected loss than the raw response is

$$\Gamma_{uc,l} \geq -\frac{w_{dd}}{w_{uc}}\Gamma_{dd,l}. \tag{9}$$

Under the maintained $\Gamma_{uc,l} > 0$, this weak-accuracy condition can fail only if $\Gamma_{dd,l} < 0$ and the weighted corrective benefit from the difference-dependent component strictly exceeds the loss generated by the unconditional component.

5.5 Absolute Accuracy: How Closely Does the RIV Approximate Genuine Valuations?

The preceding discussion asks whether removing the shared non-risk-specific mistakes makes the RIV more accurate than raw responses. A separate question is whether the RIV is close to genuine valuations in an absolute sense. Under the sequential representation in Section 3, the risk-specific stochastic structure G_l^{RS} maps the genuine valuation V_l into the RIV W_l, so that

$$\boldsymbol{w}_l = G_l^{RS}\boldsymbol{q}_l.$$

Absolute validity therefore depends on the importance of the mistakes contained in G_l^{RS}. If the risk-specific stage usually leaves genuine valuations unchanged, with G_l^{RS} concentrated near its diagonal, then most mistakes arise in the shared non-risk-specific stage G_M, and the RIV will closely track genuine valuations. If instead risk-specific mistakes are substantial, the RIV may remain far from genuine valuations even after the non-risk-specific stage has been removed.

Several theories of lottery valuation make it plausible that important sources of observed deviations are not unique to decisions under risk. Prospect Theory draws on the general psychological principles of reference dependence and diminishing sensitivity (Kahneman and Tversky, 1979, Tversky and Kahneman, 1992, Wakker, 2025). More recent theories of cognitive imprecision and efficient coding explain lottery anomalies partly through noisy representations of numerical magnitudes (Woodford, 2012, Khaw, Li and Woodford, 2021, Frydman and Jin, 2021), while salience theory invokes broader attentional mechanisms (Bordalo, Gennaioli and Shleifer, 2012). These theories do not establish that G_l^{RS} is close to the identity matrix, but they make it plausible that mechanisms shared with non-risky tasks account for an important part of the stochasticity in lottery responses.

For a rough benchmark, we turn to Martínez-Marquina, Niederle and Vespa (2019). Their experiment compares related deterministic and probabilistic contingent-reasoning problems. Moreover, behavior in both settings can be evaluated against a payoff-maximizing benchmark. Their Table 3 classifies 58.5% of subjects in the deterministic treatment and 80.3% in the probabilistic treatment as failing to follow the payoff-maximizing strategy. The error rate in the deterministic treatment is approximately 73% of the probabilistic treatment. This comparison suggests that difficulties already present without risk may account for an important part of the difficulties observed under risk, while the 21.8-percentage-point difference also indicates that the risk-specific component need not be negligible.

Our data likewise suggest that the risk-specific mistakes may retain an important distortion. Under the \$100 prize condition, lottery responses place substantially more mass on the smallest response, \$0, than mirror responses do, as shown by comparing Figure 1 and Figure 5.

Moreover, the mass at \$0 in lottery tasks is relatively stable across p. This excess mass is not removed by the mirror-estimated non-risk-specific stochastic structure and consequently appears as a spike at \$0 in the estimated RIV distributions in Figure 6. This excess mass could indicate that some subjects have an additional lottery-specific tendency to report \$0 regardless of p. Within the sequential representation, this tendency is part of G_l^{RS} and is therefore deliberately retained in the RIV.

6 Applying the Procedure to Other Data Sets

In this section, we apply the same two-step procedure for recovering RIV to other data sets and examine whether the main results generalize. Specifically, we assess whether the main findings in Section 4 and Section 5 are robust in data from Oprea (2024) and Zhang (2025). Both studies conducted experiments that collected valuations of lotteries and mirrors from the same subjects, allowing a seamless application of the two-step procedure.

Oprea (2024) To assess the robustness and generalizability of our findings, we apply the same two-step procedure to data from the main experiment in Oprea (2024). In that experiment, subjects recruited from Prolific valued lotteries and their deterministic mirrors through price lists, with a prize of \$25. The main difference from our study lies in the experimental instructions and subject screening procedures. Despite procedural differences, the data allow us to test whether our methodological findings generalize across experimental implementations. To implement the mixture modeling using Oprea's (2024) data with the greatest comparability with the current paper, we exclude from his data set the lotteries that involve losses, along with the mirrors corresponding to these excluded lotteries.

Figure 7 demonstrates that Model 1 continues to provide an excellent fit to mirror responses in Oprea's data. The simulated distributions closely track the empirical distributions across all probability levels, indicating that the three-component mixture model – with difference-dependent and unconditional mistakes alongside rational responses – captures the dominant patterns of behavior in mirror valuation tasks even under different experimental protocols. This robustness suggests that these two mistake types represent fundamental features of how subjects respond in mirror valuation tasks, rather than artifacts specific to our experimental design.

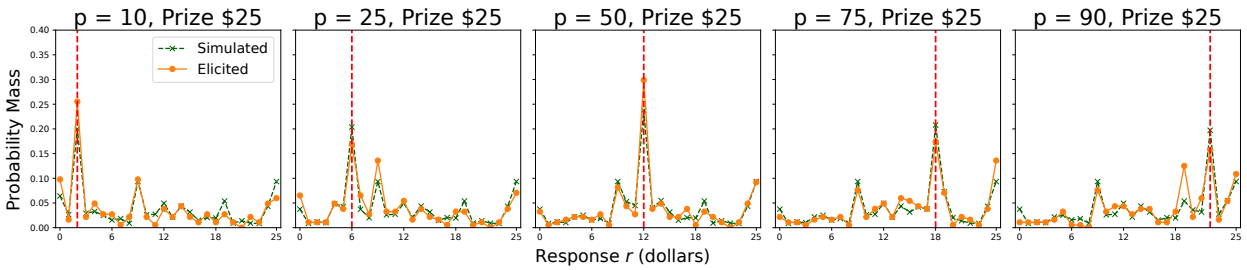


Figure 7: Model fit in mirror tasks using data from Oprea (2024)

Table 3 shows that the main substantive finding from Section 5 replicates in Oprea's data: The RIV are substantially closer to the risk-neutral benchmark than raw responses, making them more consistent with expected utility theory's prediction of approximate risk neutrality under small stakes (Rabin, 2000). This pattern is particularly striking for small-probability gains. For the lottery with $p = 10\%$, the median response is \$8.50, reflecting strong risk seeking. In contrast, the median RIV equals the risk-neutral benchmark (\$2), and the mean RIV is only moderately above it (\$3.00). This represents a dramatic shift from the pronounced risk seeking in raw responses toward risk neutrality after correcting for mistakes common to mirror and lottery tasks.

p (%)	Risk-neutral	Mean		Median		Share of RIV		
	benchmark	RIV	Responses	RIV	Responses	Risk-averse	Risk-neutral	Risk-seeking
10	2	3.00 [2.45,5.11]	8.55 [7.49,9.59]	2 [2.00,4.00]	8.50 [4.00,9.00]	0.22	0.34	0.43
25	6	5.40 [4.51,7.52]	9.47 [8.48,10.45]	6 [4.00,6.00]	9.00 [7.00,9.00]	0.40	0.33	0.27
50	12	9.47 [7.84,11.49]	11.28 [10.35,12.19]	12 [7.00,14.00]	12.00 [10.00,12.00]	0.46	0.24	0.30
75	18	13.45 [11.35,15.30]	13.55 [12.59,14.52]	14 [10.00,18.00]	14.00 [13.00,15.00]	0.67	0.27	0.06
90	22	18.12 [15.75,19.68]	15.94 [14.96,16.93]	20 [19.00,21.00]	18.00 [15.00,19.00]	0.72	0.22	0.06

Table 3: Summary statistics of risk-induced valuations and responses in the Oprea application.
Notes: Mean and median cells report point estimates above percentile 95% confidence intervals. Intervals use $B = 2{,}000$ paired subject-clustered whole-pipeline bootstrap replications. Each replication re-estimates both Model 1 and the RIV distribution; response summaries are recomputed from the same resampled subjects.

Zhang (2025) We apply the mixture modeling approach to the data set of Zhang (2025), which has a few key advantages. First, the lottery and mirror valuations in Zhang's (2025) data set were incentivized using the Becker-DeGroot-Marschak (BDM) mechanism (Becker,

Degroot and Marschak, 1964), unlike the price-list valuations in this paper. Therefore, a test of the mixture modeling on this data set expands the potential scope of the approach. Second and more importantly, Zhang's (2025) data set includes the *same set of subjects* performing the same set of lottery and mirror tasks twice, once with the help of a calculator (the *Calc* treatment) and once without (the *NoCalc* treatment). Zhang (2025) shows that elicited lottery valuations are somewhat different between the NoCalc and Calc treatments, and demonstrates that the differences are largely due to different stochastic structures in the two treatments, since the Calc treatment reduces mistakes by lowering the costs of performing explicit calculations.

The fact that Zhang's (2025) data set includes repeated elicitation of lottery and mirror valuations under different stochastic structures provides a testing ground for whether RIV removes non-risk-specific mistakes. Non-risk-specific mistakes induced by computational difficulty will differ across the NoCalc and Calc treatment, but genuine valuations and risk-specific mistakes stay the same. Thus, our theoretical framework would predict identical RIV across the two treatment despite different raw responses.

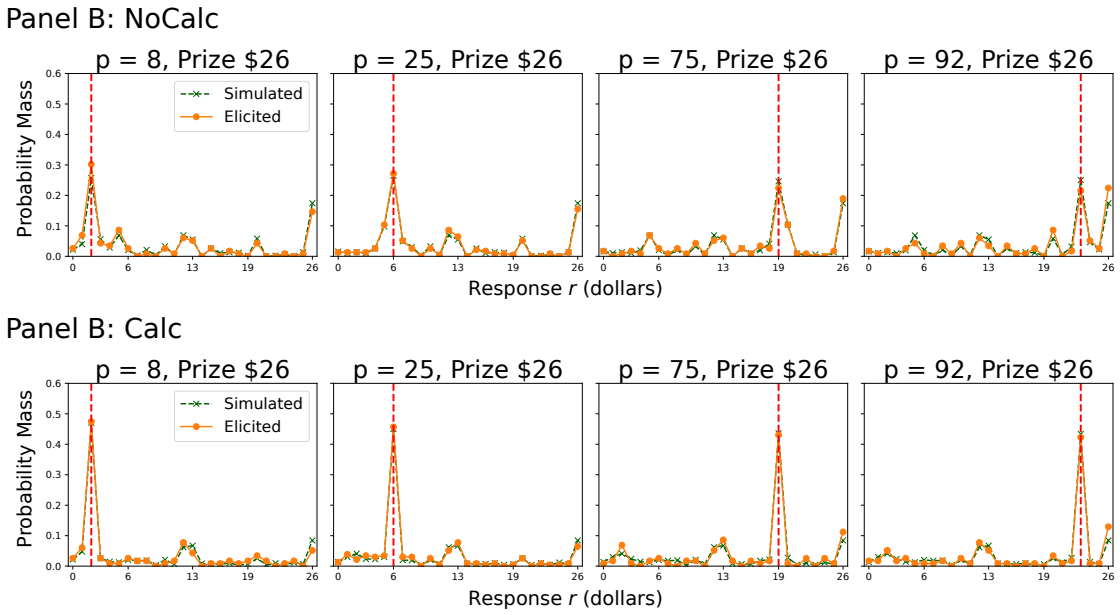


Figure 8: Model fit in mirror tasks using data from Zhang (2025)

To maximize the comparability with the current study, we only use lotteries involving gains from Zhang (2025), while dropping any lotteries involving losses. Moreover, since the valuations under the BDM mechanism can take a continuum of possible values, we need to discretize the BDM valuations, in order to make them suitable for estimating the mixture model parameters. To achieve this, we round down the BDM valuations to integers and generate the "responses." The RIV distributions are then recovered by separately using the

valuation data from each of the NoCalc and Calc treatments.

Figure 8 shows the fit of Model 1 using Zhang's (2025) mirror valuation data, by juxtaposing the empirical distributions of responses with the distributions simulated from the estimated mixture model, separately in the two treatments. Visibly, Model 1 again provides a great fit for the empirical distributions, which shows that the three-component decomposition of mirror responses generalizes to another data set collected using a different method.

Panel A of Figure 9 shows the cumulative distribution functions (CDFs) of the RIV distributions using data from Zhang (2025). In both the NoCalc and Calc treatments, the RIV distributions are highly concentrated around the risk-neutral benchmarks of the lotteries. For every lottery-treatment combination, the RIV distribution assigns more than 50% probability mass exactly to the risk-neutral benchmark of the lottery. For comparison with the RIV distributions in Panel A, Panel B of Figure 9 juxtaposes the CDFs of the responses in the NoCalc and Calc treatments. The RIV distributions are much more similar across the NoCalc and Calc treatments than the raw-response distributions are. More specifically, for the responses, the Calc treatment has much larger probability masses at the risk-neutral benchmark of the lottery than the NoCalc treatment. In contrast, for the RIV, the two treatments have more similar probability masses at the risk-neutral benchmark.

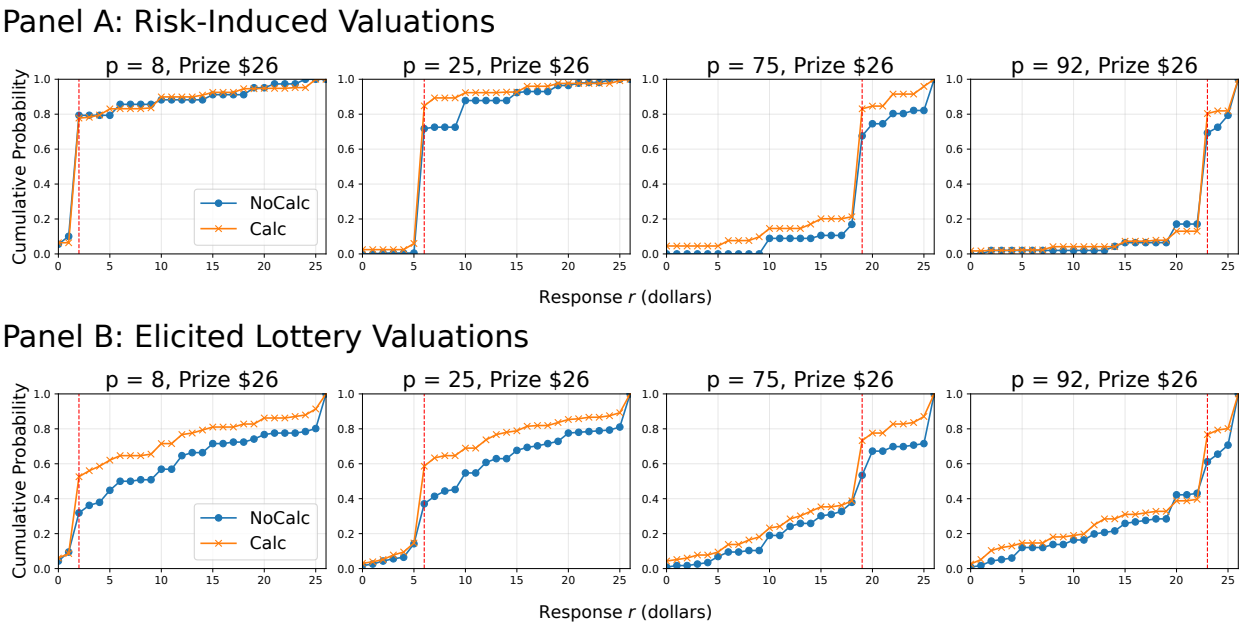


Figure 9: The cumulative distribution function of elicited lottery valuations and recovered risk-induced valuations in Zhang (2025)

Next, we test two joint null hypotheses for every retained lottery l: equal distributions of RIV, $H_0^W : \boldsymbol{w}_l^{\text{NoCalc}} = \boldsymbol{w}_l^{\text{Calc}}$, and equal distributions of responses, $H_0^R : \boldsymbol{p}_l^{\text{NoCalc}} = \boldsymbol{p}_l^{\text{Calc}}$. For each hypothesis, the test statistic is the maximum absolute difference between the Calc

and NoCalc CDFs across all retained lotteries and all nonredundant response thresholds. We obtain p-values from a centered, subject-clustered bootstrap that resamples complete subject records and re-estimates the full correction procedure in every replication. Appendix B provides the details and formal justification. With 2,000 bootstrap replications, the test does not reject equal RIV distributions ($p = 0.53$), while it rejects equal response distributions ($p < 0.01$). Thus, this omnibus test detects the treatment difference in raw responses but not in the RIV after correction. This result passes a necessary-condition check for interpreting RIV as an approximation to genuine valuations. It is consistent with successful correction of treatment-specific, non-risk-specific mistakes, but does not by itself establish equality, equivalence, or the absence of residual mistakes.

7 Conclusion

This paper contributes to our understanding of risk preferences by distinguishing between genuine preferences and mistakes in experimental responses. By correcting for mistakes that are common to both risky and risk-free decision contexts – using deterministic mirror tasks where correct valuations are known – we find that subjects' valuations are substantially closer to risk neutrality than raw responses suggest. When risk-induced valuations deviate from risk-neutral benchmarks, they predominantly exhibit risk aversion rather than the pronounced risk seeking for low-probability gains documented in prior work. These findings suggest that the literature may have overestimated the extent to which risk preferences deviate from expected utility theory, and that estimates of prospect-theoretic parameters – particularly the strong overweighting of small probabilities – should not be interpreted as solely reflecting genuine preferences, but as also reflecting systematic mistakes in responding to preference elicitation tasks.

Beyond these substantive findings, this paper's primary contribution may be methodological. We develop a two-step procedure that reduces reliance on unverifiable parametric assumptions about stochastic choice by estimating the structure of mistakes in a related context where they are observable. The key innovation is the pipeline involving deterministic mirrors – objects with known valuations that share important features with the target decision problem. In our setting, both lotteries and their mirrors require processing the same disaggregated visual information and navigating the same price list mechanism, allowing us to identify mistake patterns that plausibly carry over to lottery tasks. By estimating a mixture model that characterizes these patterns, we obtain an empirically grounded correction for lottery responses that avoids arbitrary functional form assumptions. In the validation exercise using Zhang (2025), the equality test rejects for raw responses but not for recovered RIV after a treatment changes non-risk-specific mistake rates. This evidence is consistent with

the correction operating as intended.

This approach may prove useful beyond the study of risk preferences. Whenever researchers seek to recover preferences from observed choices, they face the challenge that choices reflect both preferences and mistakes. Our method suggests a path forward: identify an auxiliary decision problem – a "mirror" – that shares key features with the target decision context, particularly those features likely to generate mistakes. The spirit of the mirror design is to match the cognitive demands of the original task; if disaggregation is a likely source of mistakes, the mirror should involve similar disaggregation; if the incentivizing mechanism, like price lists or the BDM mechanism, is likely to cause confusion, the mirror should be elicited with the same mechanism. Researchers can then pursue one of two strategies. If mistakes in the mirror setting are negligible or cancel out on average, or if the statistical method employed by the researchers correctly recovers preferences despite mistakes, the mirror serves as a validation device without requiring explicit correction. However, if mistakes in the mirror would lead to erroneous conclusions about preferences, then a correction procedure similar to our two-step approach becomes necessary: estimate the mistake structure in the mirror setting, then apply it to correct choices in the target domain. Researchers adopting this correction approach must be explicit about what their corrected measure captures and what it does not. In our case, the risk-induced valuations capture both genuine risk preferences and any risk-specific mistakes not present in the deterministic mirror tasks, but cannot be interpreted as genuine preferences without additional assumptions. Any application of this methodology requires similar careful delineation of what has been corrected for (mistakes common to both contexts) and what remains (domain-specific mistakes), along with transparent discussion of the assumptions required for the correction to recover genuine preferences. While substantial work remains to understand when and how mistake structures transfer across contexts, this paper demonstrates that empirically disciplined approaches to correcting for mistakes are feasible and can materially change our conclusions about preferences.

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Appendices

A Out-of-Sample Predictive Power of Model 1

The mixture model fits the in-sample conditional probabilities $\{g_M(r \mid v) : r \in \mathcal{R}, v \in V_{\mathcal{M}}\}$ well. How well does it predict the out-of-sample conditional probabilities $\{g_M(r \mid v) : r \in \mathcal{R}, v \in V_{\mathcal{M}}^c\}$? Borrowing techniques from machine learning, we now assess the out-of-sample predictive power of the mixture model and compare it with several benchmark specifications of the stochastic structure.

The in-sample fit of the mixture model is not suitable for assessing its out-of-sample predictive power, since the model is estimated using exactly the same data set that the model is assessed on (see, e.g., Hastie, Tibshirani and Friedman, 2009, Section 7.4). Instead, to assess the out-of-sample predictive power, we need to use a different data set than the one on which the model parameters are estimated. To this end, we employ the algorithm of leave-one-m-out cross-validation. This algorithm works by first excluding all observations involving a fixed mirror m from D_M and generating a sub-data set $D_{-m}^{\text{estimation}}$. Next, the procedure estimates the parameters of the mixture model using the new data set $D_{-m}^{\text{estimation}}$ and generates parameter estimates $\hat{\Psi}_{-m}$. Then, the procedure evaluates the parameter estimates $\hat{\Psi}_{-m}$ using the sub-data set $D_m^{\text{validation}}$ that includes all observations involving the mirror m. Let $\mathcal{L}(\hat{\Psi}_{-m}; D_m^{\text{validation}})$ denote the log likelihood on that validation fold. Therefore, the data set that the model is evaluated on, $D_m^{\text{validation}}$, is entirely different from the data set on which the model is estimated, $D_{-m}^{\text{estimation}}$. Finally, the algorithm repeats the process above for all $m \in \mathcal{M}$ and uses the average log likelihood

$$\bar{\mathcal{L}} := \frac{1}{|\mathcal{M}|} \sum_{m \in \mathcal{M}} \frac{1}{|D_m^{\text{validation}}|} \mathcal{L}(\hat{\Psi}_{-m}; D_m^{\text{validation}}) \tag{10}$$

as the measure of the out-of-sample *predictive power*. Similar cross-validation procedures are widely adopted in machine learning in evaluating the out-of-sample performance of predictive models (Hastie, Tibshirani and Friedman, 2009, Section 7.10), and have been increasingly adopted in economics (see, e.g., Kleinberg et al., 2018, Fudenberg et al., 2022, Mullainathan and Obermeyer, 2022).

The predictive power $\bar{\mathcal{L}}$ is a relative measure that only becomes meaningful when compared between different models. To this end, we also construct the same measure, $\bar{\mathcal{L}}$, for a few benchmark specifications of the stochastic structure $g_M(r \mid v)$ for comparison. For each of these benchmark specifications, we first estimate its parameters, if any, and then compute the predictive power $\bar{\mathcal{L}}$ through the same leave-one-m-out cross-validation explained above. The benchmark specifications include the following.

- **Uniform Choice Lower Bound** assumes uniform choice of responses conditional on any v: $g_M(r \mid v) = \frac{1}{|\mathcal{R}|}$. This "guess the choice at random" model can be viewed as a "lower bound" on $\bar{\mathcal{L}}$ since any reasonable prediction should not be worse than uniform choice.

- **Omniscient Upper Bound** uses the empirical distribution of r conditional on v observed in the data to "predict" itself:

$$g_{M,\text{empirical}}(r \mid v) := \frac{\sum_{(i,m,r_{m,i}) \in D_M} \mathbb{1}[r_{m,i} = r,\ v(m) = v]}{\sum_{(i,m,r_{m,i}) \in D_M} \mathbb{1}[v(m) = v]}.$$

  Under this benchmark, the predictive power $\bar{\mathcal{L}}$ achieves the theoretical upper bound for the data set D_M. This bound is unattainable in practice but provides a natural upper scale for $\bar{\mathcal{L}}$.

- **Uniform Mistake** assumes that with probability $1-\epsilon$, the subjects choose the genuine valuation, and with the remaining probability ϵ, they make a mistake by choosing responses uniformly in $\mathcal{R}$. The set of parameters in this stochastic structure is $\{\epsilon\}$. This stochastic structure is a generalization of Harless and Camerer (1994).

- **Normal Mistake** assumes a discrete Gaussian response distribution centered on the genuine valuation and normalized over the finite response grid $\mathcal{R}$. Its variance is heterogeneous across individual subjects. The set of parameters is $\{\sigma_i$ for each subject $i\}$. The stochastic structure is a generalization of Hey and Orme (1994) that is adopted by Bruhin, Fehr-Duda and Epper (2010).[13]

We also include two restricted versions of Model 1 in the comparison. These models are described in terms of restrictions we impose on the parameters Ψ, but otherwise they are identical to Model 1.

- **Uniform Unconditional Mistake** restricts the unconditional mistake in Model 1 to be a uniform distribution over all $r \in \mathcal{R}$: $\lambda_r = \frac{1}{|\mathcal{R}|}$ for all $r \in \mathcal{R}$. This differs from the Uniform Mistake above by allowing for difference-dependent mistakes.

[13]When implementing the normal stochastic structure, we impose a restriction that $\sigma_i > \underline{\sigma}$ for all i, where $\underline{\sigma}$ is a lower bound. The rationale can be explained by an example. Suppose that a subject i chooses $r_{m,i} \neq v(m)$ for m, but $r_{m',i} = v(m')$ for all other $m' \neq m$. Without the lower bound on σ_i, the $\sigma_{i,-m}$ estimated using $D_{-m}^{\text{estimation}}$ will be 0. However, with $\sigma_{i,-m} = 0$, the likelihood of the observation $(i, m, r_{m,i})$ is zero and the log likelihood is negative infinity. Therefore, by imposing a lower bound for σ_i, we avoid this scenario and get better predictive power. The lower bound $\underline{\sigma}$ is chosen by maximizing the predictive power $\bar{\mathcal{L}}$. Because the same cross-validation criterion is used both to choose this bound and to report the Normal Mistake score, the comparison is favorable to that benchmark.

- **Unconditional Mistake Only**: imposes $w_{dd} = 0$ in Model 1, thereby excluding difference-dependent mistakes.

	Prize \$25	**Prize \$100**
Baseline Mixture Model 1	**0.664**	**0.891**
Uniform Choice Lower Bound	0	0
Omniscient Upper Bound	1	1
Uniform Mistake	0.463	0.832
Normal Mistake	0.289	0.388
Uniform Unconditional Mistake	0.639	0.867
Unconditional Mistake Only	0.337	0.494

Table A.1: The normalized predictive power for each specification of stochastic structure

Appendix Table A.1 shows the predictive power of each specification of stochastic structure. For a specification s, we report the normalized score

$$\widetilde{\mathcal{L}}_s := \frac{\bar{\mathcal{L}}_s - \bar{\mathcal{L}}_{\text{uniform}}}{\bar{\mathcal{L}}_{\text{empirical}} - \bar{\mathcal{L}}_{\text{uniform}}},$$

so that the Omniscient Upper Bound has a score of 1 and the Uniform Choice Lower Bound has a score of 0.[14] Three main takeaways arise from Appendix Table A.1. First, the mixture model has better predictive power than the stochastic structures of Uniform Mistake and Normal Mistake commonly assumed in the literature. Second, comparing the mixture model with its two restricted versions, the mixture model has better predictive power than the Uniform Unconditional Mistake specification, and much better than the Unconditional Mistake Only specification. In these comparisons, allowing both difference-dependent mistakes and a flexible unconditional mistake distribution improves predictive performance.[15] Third, the mixture model performs best among the feasible specifications compared here, and its score is especially close to the Omniscient Upper Bound under prize \$100.

[14]The notion of normalized predictive power is related to the notion of *completeness* in Fudenberg et al. (2022). However, the key distinction is that the object we predict is a *distribution* of values, while most applications of Fudenberg et al. (2022) pertain to providing a *point prediction*. The fact that we predict distributions makes it very hard for us to define and estimate the "irreducible noise" as defined in Fudenberg et al. (2022), which is a key component in measuring completeness. In the meantime, the normalized predictive power can be seen as a lower bound on completeness, because any definition of irreducible noise will yield a completeness measure higher than the normalized predictive power.

[15]A "Difference-Dependent Mistake Only" specification is not possible, since the combination of the rational distribution and the difference-dependent mistake only allows responses within the grid window from $v - KS$ to $v + KS$. However, the empirical support of r lies outside this window, and without a component distribution that allows positive probability for every $r \in \mathcal{R}$, the log likelihood approaches negative infinity.

B Bootstrap Tests in the Zhang Application

The data from Zhang (2025) form a within-subject panel: the same $N = 116$ subjects complete mirror and lottery tasks in both the NoCalc and Calc treatments. We therefore treat a subject, rather than an individual response, as the independent sampling unit. We retain four gain-lottery distributions, indexed by win probabilities 8%, 25%, 75%, and 92%. The two repetitions of the 25% lottery are pooled when estimating its distribution, but remain part of the same subject record for inference.

Hypotheses and test statistics. Let $t \in \{\text{NoCalc}, \text{Calc}\}$. For each retained lottery l and $x \in \mathcal{R}$, define the estimated CDFs

$$\hat{F}^t_{W,l}(x) := \sum_{\substack{y \in \mathcal{R} \\ y \le x}} \hat{w}^t_{l,y}, \qquad \hat{F}^t_{R,l}(x) := \sum_{\substack{y \in \mathcal{R} \\ y \le x}} \hat{p}^t_l(y).$$

The joint null hypotheses are

$$\begin{aligned} H_0^W &: \boldsymbol{w}_l^{\text{NoCalc}} = \boldsymbol{w}_l^{\text{Calc}} \quad \text{for every retained lottery } l, \\ H_0^R &: \boldsymbol{p}_l^{\text{NoCalc}} = \boldsymbol{p}_l^{\text{Calc}} \quad \text{for every retained lottery } l. \end{aligned}$$

Define the Calc-minus-NoCalc RIV-CDF contrast by

$$\hat{\Delta}_{W,l}(x) := \hat{F}^{\text{Calc}}_{W,l}(x) - \hat{F}^{\text{NoCalc}}_{W,l}(x),$$

and define $\hat{\Delta}_{R,l}(x)$ analogously for responses. The two statistics are

$$\begin{aligned} T_W &:= \sqrt{N} \max_l \max_{x \in \mathcal{R} \setminus \{\max \mathcal{R}\}} \left| \hat{\Delta}_{W,l}(x) \right|, \\ T_R &:= \sqrt{N} \max_l \max_{x \in \mathcal{R} \setminus \{\max \mathcal{R}\}} \left| \hat{\Delta}_{R,l}(x) \right|. \end{aligned}$$

The highest grid value is omitted because both CDFs equal one there. On the response grid used in this application, each statistic therefore takes a maximum over $4 \times 26 = 104$ coordinates. This single maximum tests the distributional equality restrictions jointly rather than conducting 104 separate tests.

Bootstrap procedure. For each replication $b = 1, \ldots, B$, we draw N subjects with replacement. Whenever a subject is drawn, we retain all of that subject's observations from both treatments and both tasks, including both repetitions of the 25% lottery. This common draw preserves the pairing across Calc and NoCalc and the dependence across observations

within a subject (MacKinnon, Nielsen and Webb, 2023).

For the RIV test, we then repeat the complete two-step procedure within the bootstrap sample. We separately re-estimate Model 1 in the NoCalc and Calc treatments, construct each treatment's $\hat{G}_M^t$, and recover each $\hat{\boldsymbol{w}}_l^t$ using Equation (8). Thus, the null does not require the two treatments to share the same stochastic structure, and uncertainty from estimating the mirror stochastic structures enters the bootstrap distribution. For the response test, we calculate the empirical response CDFs from the same subject resample.

Let $\hat{\boldsymbol{\Delta}}_W$ stack the 104 RIV-CDF contrasts and let $\hat{\boldsymbol{\Delta}}_W^{*(b)}$ denote their bootstrap analogues. The centered bootstrap statistic is

$$T_W^{*(b)} := \sqrt{N} \left\| \hat{\boldsymbol{\Delta}}_W^{*(b)} - \hat{\boldsymbol{\Delta}}_W \right\|_\infty ,$$

with $T_R^{*(b)}$ defined analogously. The bootstrap samples are not themselves forced to satisfy the equality null. Instead, subtracting $\hat{\boldsymbol{\Delta}}_W$ makes the bootstrap distribution approximate the sampling error of the estimated contrast. Under H_0^W, the population contrast is zero, so the observed $\sqrt{N}\hat{\boldsymbol{\Delta}}_W$ is itself sampling error. Without this subtraction, the bootstrap distribution would remain centered near the observed treatment difference and would not estimate the desired null reference distribution.

This argument can be stated more formally. Let $\boldsymbol{\Delta}_W$ denote the population vector of CDF contrasts. Suppose subjects are independent sampling clusters, the number of observations per subject is bounded, the population criteria locally identify the treatment-specific mirror and RIV estimators, and the map from the subject-level empirical distribution through both estimation stages to $\hat{\boldsymbol{\Delta}}_W$ is Hadamard differentiable. Under these regularity conditions,

$$\sqrt{N}\left(\hat{\boldsymbol{\Delta}}_W - \boldsymbol{\Delta}_W\right) \rightsquigarrow \mathbb{G}_W,$$

and, conditional on the observed data, the distribution of $\sqrt{N}(\hat{\boldsymbol{\Delta}}_W^* - \hat{\boldsymbol{\Delta}}_W)$ converges in probability to the same limiting law. Under H_0^W, $\boldsymbol{\Delta}_W = \mathbf{0}$. Because the maximum absolute value is a continuous function of this finite-dimensional vector, the centered bootstrap consistently estimates the null distribution of T_W whenever the limiting distribution is continuous at the relevant critical value. The same reasoning applies directly to T_R.

For either $T \in \{T_W, T_R\}$, we use the finite-simulation corrected p-value

$$\hat{p} = \frac{1 + \sum_{b=1}^{B} \mathbb{1}[T^{*(b)} \geq T]}{B+1}.$$

Results. With $B = 2{,}000$, the RIV statistic is $T_W = 1.8049$ and 1,062 bootstrap statistics are at least this large, giving $p_W = 0.5312$. The response statistic is $T_R = 2.3676$ and none of the 2,000 bootstrap statistics is as large, giving the finite-simulation value $p_R = 1/2{,}001 = 0.0005$. The main text reports these results as $p = 0.53$ and $p < 0.01$, respectively.

Boundary sensitivity. The regularity argument above may fail if population RIV probabilities lie on the boundary of the probability simplex, because the constrained estimator need not then be differentiable in the usual way (Andrews, 2000). Some fitted probabilities are zero or very close to zero, so we supplement the ordinary bootstrap with an m-out-of-n sensitivity analysis. To avoid confusing the resample size with the mirror index m, denote the smaller resample size by N_s. For each N_s, we draw N_s complete subject records with replacement, repeat both estimation stages, and use

$$T^{*(b)}_{W,N_s} := \sqrt{N_s}\left\|\hat{\boldsymbol{\Delta}}^{*(b)}_{W,N_s} - \hat{\boldsymbol{\Delta}}_W\right\|_\infty$$

as the reference statistic for the full-sample T_W. The formal m-out-of-n argument considers sequences with $N_s \to \infty$ and $N_s/N \to 0$ (Bickel, Götze and van Zwet, 1997). With the present N, finite choices of N_s are therefore a sensitivity check rather than proof of uniform validity. Using 2,000 replications at each of $N_s \in \{50, 70, 90\}$ gives RIV p-values of 0.43, 0.50, and 0.53, respectively. The non-rejection conclusion is unchanged across these choices.

C Experimental Instructions

In this study, you will face 10 choice tasks. At the end of the study, if you are randomly selected to receive a bonus payment, then we will randomly select one of the 10 tasks, and we will determine your bonus payment based on your responses in this one randomly selected task.

Note: Each of the decisions you make today may determine your bonus payment. Therefore, it is in your best interest to answer every decision carefully and in a way that reflects what you'd truly prefer.

Next

Boxes paying out money

All choice tasks in this study will involve a set of boxes paying out money according to a payment rule.

Boxes paying out money

You will be facing 100 boxes that pay out money according to a payment rule.

Example:

30 Boxes	70 Boxes
$100	$0

Payment rule: You will receive the **average amount of money** in the set of boxes **with certainty**. In this example, you would receive $30, obtained by ($100*30+$0*70)/100 = $30.

Next

Three simulations are provided.

Simulated Payment from the Boxes

Example:

75 Boxes	25 Boxes
$100	$0

To help you better understand the possible outcomes, we simulate 10 random outcomes from the boxes above.

- $75
- $75
- $75
- $75
- $75
- $75
- $75
- $75
- $75
- $75

Next

Next Step: Comprehension Check

You'll be asked a few questions to make sure you understand the task. To pass the check, you need to answer 4 out of 5 questions correctly.

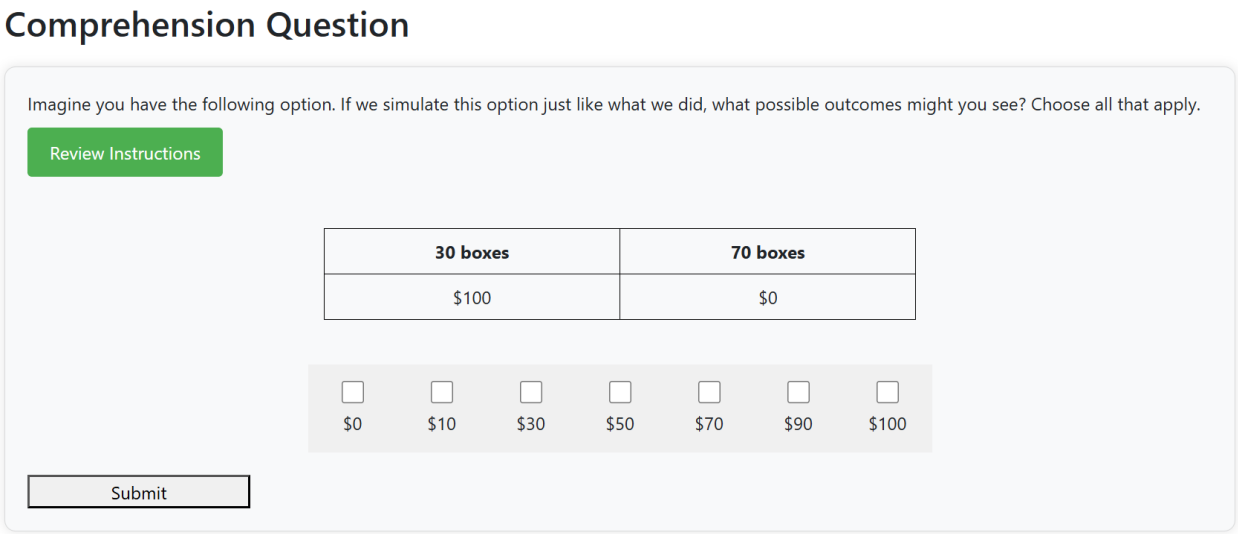
If you answer fewer than 4 questions correctly, you will be screened out of the study. You will not be paid for your participation in this case.

If you need to review the instructions, a link will be available for quick access.

Feel free to proceed when you're ready!

Next

Five comprehension questions are asked.



Choice Tasks

Each choice task will involve one or several decisions between a set of boxes called Option A and a certain payment called Option B. For each decision, all you have to do is decide whether you prefer Option A or Option B.

Your decisions in the choice tasks will be presented in a table in which each row represents a separate decision. In the example that you can find in the bottom of this page, the decision in row 1 involves the choice between Option A (getting paid according to the boxes) and Option B (a certain payment of **$0**). The decision in row 2 involves a choice between the same Option A, but now Option B gives a 100% chance of receiving **$1**. In row 3, Option B gives a 100% chance of receiving **$2**, and so on.

The amount of money in Option B will change in each row. We will put the money amount in **bold** to help you see what changes.

Option A will stay the same in each row of a given choice task.

Example of a choice task

Option A is being paid according to the boxes:

100 Boxes	0 Boxes
$10.50	$0

Decision	Option A	Option B
1	Being paid according to the boxes	**$0.00** for sure
2	Being paid according to the boxes	**$1.00** for sure
3	Being paid according to the boxes	**$2.00** for sure
4	Being paid according to the boxes	**$3.00** for sure
5	Being paid according to the boxes	**$4.00** for sure
6	Being paid according to the boxes	**$5.00** for sure
7	Being paid according to the boxes	**$6.00** for sure
8	Being paid according to the boxes	**$7.00** for sure
9	Being paid according to the boxes	**$8.00** for sure
10	Being paid according to the boxes	**$9.00** for sure
11	Being paid according to the boxes	**$10.00** for sure
12	Being paid according to the boxes	**$11.00** for sure
13	Being paid according to the boxes	**$12.00** for sure
14	Being paid according to the boxes	**$13.00** for sure
15	Being paid according to the boxes	**$14.00** for sure
16	Being paid according to the boxes	**$15.00** for sure

Next

Choice Tasks: Example

To give you an example of how these choice tasks work, consider the hypothetical task below. Option A is 100 boxes, each paying out $10.50. Option B gives a sure payment that varies from $0 (in the first row) to $15 (in the last row).

In this hypothetical decision, if you prefer more money to less, you should prefer Option A (paying out $10.50 for sure) for the first 11 rows. Starting from row 12 (100% chance of $11), you should prefer Option B.

To avoid having to make a selection in each row, you need only click two times: once for Option A and once for Option B. Once you click on Option A in a particular row, all rows above that row will populate to indicate that you choose Option A. Once you click a row in Option B, all rows below that row will populate to indicate that you choose Option B.

In the example below, in the 11th row, you would click Option A (being paid according to the boxes), instead of Option B ("$10 for sure"). In contrast, in the 12th row, you would click Option B ("$11 for sure") instead of Option A.

If you do what is described above, your choices will be as in the table below.

IMPORTANT: In order to progress, you have to make a selection for every row.

An hypothetical example of the choices you will be making

Option A is being paid according to the boxes:

100 Boxes	0 Boxes
$10.50	$0

Your choice will be the following if you prefer more money to less:

Decision	Option A	Option B
1	Being paid according to the boxes	**$0.00** for sure
2	Being paid according to the boxes	**$1.00** for sure
3	Being paid according to the boxes	**$2.00** for sure
4	Being paid according to the boxes	**$3.00** for sure
5	Being paid according to the boxes	**$4.00** for sure
6	Being paid according to the boxes	**$5.00** for sure
7	Being paid according to the boxes	**$6.00** for sure
8	Being paid according to the boxes	**$7.00** for sure
9	Being paid according to the boxes	**$8.00** for sure
10	Being paid according to the boxes	**$9.00** for sure
11	Being paid according to the boxes	**$10.00** for sure
12	Being paid according to the boxes	**$11.00** for sure
13	Being paid according to the boxes	**$12.00** for sure
14	Being paid according to the boxes	**$13.00** for sure
15	Being paid according to the boxes	**$14.00** for sure
16	Being paid according to the boxes	**$15.00** for sure

Next

Choice Tasks: Practice

To make sure you are comfortable with how the choice tables work, please complete the following practice choice.

1. How to select Option A in all rows
2. How to select Option B in all rows
3. How to change your answer after selecting

As in the main decisions, you will not be able to click to the next page until you have made a valid selection. To make a valid selection, you have to make a selection for every row. That is, your choices must either "switch" from Option A to Option B (clicking once on a row in Option A and once on the subsequent row in Option B to proceed), or you can click Option A in all rows, or Option B in all rows.

If the page is not letting you proceed, you have not made a valid selection. Please make sure you click on Option A in the last row where you would want to select Option A, and then click on Option B in the first row you would want to select Option B (the subsequent row after selecting Option A).

Decision	Option A	Option B
1	Being paid according to the boxes	**$0.00** for sure
2	Being paid according to the boxes	**$1.00** for sure
3	Being paid according to the boxes	**$2.00** for sure
4	Being paid according to the boxes	**$3.00** for sure
5	Being paid according to the boxes	**$4.00** for sure
6	Being paid according to the boxes	**$5.00** for sure
7	Being paid according to the boxes	**$6.00** for sure
8	Being paid according to the boxes	**$7.00** for sure
9	Being paid according to the boxes	**$8.00** for sure
10	Being paid according to the boxes	**$9.00** for sure
11	Being paid according to the boxes	**$10.00** for sure
12	Being paid according to the boxes	**$11.00** for sure
13	Being paid according to the boxes	**$12.00** for sure
14	Being paid according to the boxes	**$13.00** for sure
15	Being paid according to the boxes	**$14.00** for sure
16	Being paid according to the boxes	**$15.00** for sure

Note that you can still change your selection once the table has been auto-populated. To do so, simply click the new row in which you would want to start choosing Option B. Once you have made your final decision, you will be able to proceed to your next choice.

Confirm Choices

Payment

Your bonus payment

After you complete all 10 choice tasks, you will get a participation fee of $3.00. In addition, you may receive a bonus payment based on your choices. Here is how we will determine your bonus payment.

First, the computer will randomly select a number between 1 and 100, with each number equally likely. If the number drawn is a number between 1 and 4 (inclusive), you have been chosen to receive a possible bonus payment. In other words, you have 4% chance of receiving a bonus payment.

If you are chosen, the computer will randomly select a **choice task** that determines your payment. Then, the computer will randomly select a **decision** from that choice task (one of the rows) by drawing another random number, again with each row equally likely. This will be the **decision-that-counts**. We will determine your payment based on whatever option you chose in the decision-that-counts.

An example

For example, imagine that you had filled out the decision screen below as indicated in the bottom of the screen, and imagine further that the decision-that-counts turned out to be the third row of this decision screen (being paid according to the boxes vs. $40 for sure).

In this decision, you chose Option A (being paid according to the boxes) over Option B ($40 for sure). In this case, you would be paid according to the boxes if you are one of the randomly selected participants who will receive a bonus payment. Review How Boxes Work

If instead you had picked Option B, you will be paid $40 for sure.

Note, these bonus payments are in addition to your completion payment.

Your Hypothetical Choices

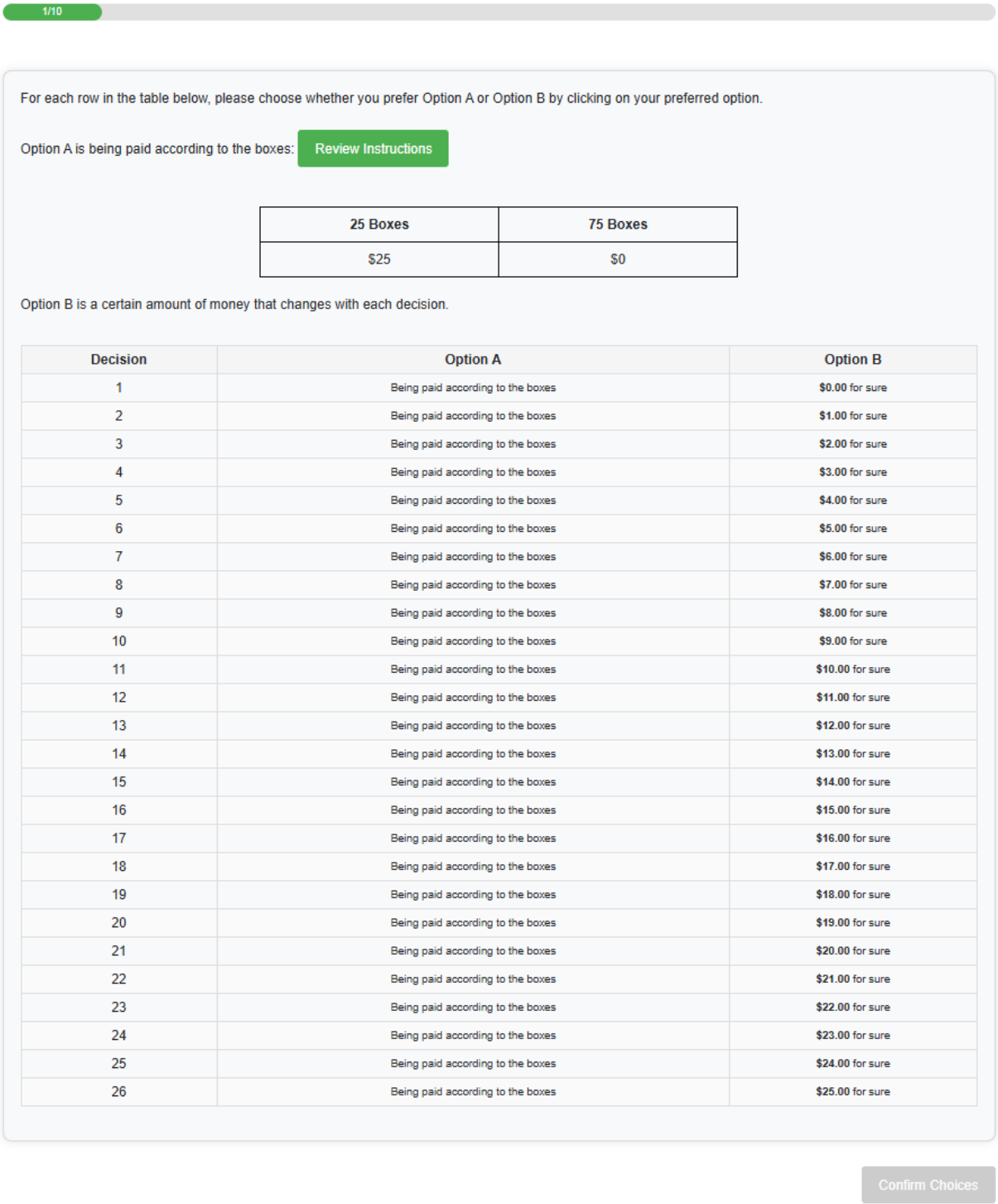
Option A is being paid according to the boxes:

50 Boxes	50 Boxes
$100	$0

Decision	Option A	Option B
1	Being paid according to the boxes	**$0.00** for sure
2	Being paid according to the boxes	**$20.00** for sure
3	Being paid according to the boxes	**$40.00** for sure
4	Being paid according to the boxes	**$60.00** for sure
5	Being paid according to the boxes	**$80.00** for sure
6	Being paid according to the boxes	**$100.00** for sure

Next

The main choice-task interface is shown below.



After five tasks, subjects begin the second payment-rule block and are shown the following notice.

Boxes paying out money

In the upcoming choice tasks, the **payment rule has changed**. Please review the new payment rule below.

Boxes paying out money

You will be facing 100 boxes that pay out money according to a payment rule.

Example:

30 Boxes	70 Boxes
$100	$0

Payment rule: The computer will calculate the outcome by **randomly picking** one of the boxes. The amount of money in that box (in this example, could be $0 or $100) will be your final result.

Next

The simulations and comprehension checks appear again for the new payment rule.